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"High Dimensional Canonical Correlation Analysis: Applications to Financial Data"

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SUMMARY

Canonical correlation analysis (CCA) measures dependence between two panels by finding the linear combinations of each panel that are maximally correlated. This thesis studies whether high-dimensional CCA (HDCCA) provides a useful framework for detecting co-movement between cyclical and defensive stock-return panels. Building on the theoretical results of Bykhovskaya and Gorin (2023), the analysis combines Monte Carlo simulations with an empirical application to stock panel data. The simulations show that large in-sample canonical roots can arise from high-dimensional noise and that reliable interpretation requires comparison with finite-sample benchmarks, directional stability, and out-of-sample performance. In the empirical application, the leading canonical roots are large relative to the HDCCA bulk benchmark and the permutation null, indicating strong spectral dependence between the two panels. However, the associated canonical directions are unstable, weakly separated from nearby roots, and perform poorly out of sample. Overall, the evidence suggests that HDCCA is useful as a diagnostic framework for detecting cross-sector dependence, but that in-sample spectral strength alone is not sufficient to establish a stable or economically meaningful low-rank structure.

Keywords: High-dimensional CCA; canonical correlation analysis; random matrix theory; financial returns; Monte Carlo simulation; CRSP.

RESUMEN

El análisis de correlación canónica (CCA) mide la dependencia entre dos paneles mediante la identificación de las combinaciones lineales de cada panel que están máximamente correlacionadas. Esta tesis estudia si el CCA en alta dimensión (HDCCA) proporciona un marco útil para detectar co-movimientos entre paneles de acciones cíclicas y defensivas. A partir de los resultados teóricos de Bykhovskaya and Gorin (2023), el análisis combina simulaciones de Monte Carlo con una aplicación empírica a datos de panel de rendimientos accionarios. Las simulaciones muestran que raíces canónicas muestrales elevadas pueden surgir por ruido de alta dimensión y que su interpretación requiere compararlas con referencias de muestra finita, estabilidad direccional y desempeño fuera de muestra. En la aplicación empírica, las raíces canónicas principales son elevadas respecto del borde del bulk de HDCCA y del nulo por permutación, lo que indica una fuerte dependencia espectral entre ambos paneles. Sin embargo, las direcciones canónicas asociadas son inestables, están débilmente separadas de raíces cercanas y muestran bajo desempeño fuera de muestra. En conjunto, la evidencia sugiere que HDCCA es útil como marco diagnóstico para detectar dependencia entre sectores, pero que la fuerza espectral dentro de muestra no basta para establecer una estructura de bajo rango estable o económicamente significativa.

Palabras clave: CCA de alta dimensión; análisis de correlación canónica; teoría de matrices aleatorias; rendimientos financieros; simulación de Monte Carlo; CRSP.

DEDICATION

To my parents, for always encouraging me to aim higher.

To my friends, for accompanying me throughout these years.

And to Carlos, for your unconditional support and trust in me.

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1. INTRODUCTION

"Sir, I will not say but that it is the part of a wise man to keep himself to-day for to-morrow, and not venture all his eggs in one basket."

What was true for farmers half a millennium ago continues to be true nowadays for financial investors. They have changed eggs for investments and baskets for portfolios, but the lesson endures: in the face of uncertainty, it is often safer not to rely on a single plan. Like eggs in a basket, some assets tend to fall at the same time, so the true choice lies in holding investments that do not all depend on the same sources of risk. This is called *diversification*, to spread investments across different opportunities, reducing dependence on any one investment; and it is essentially the theoretical financial equivalent to *Don't Put All Your Eggs in One Basket*. Effective diversification therefore requires attention to the relationships between investments, whether they tend to move together, move differently, and how they respond to changing conditions. For this reason, the statistical problem relies both on the number of assets held and on the dependence structure among their returns.

One important source of changing conditions is the business cycle, the pattern of expansion and slowdown that economies experience over time. During periods of economic growth, some companies may benefit from rising demand and increased spending. During downturns, others may prove more resilient because they provide goods or services that people continue to need.

This distinction gives rise to the contrast between cyclical and defensive stocks. *Cyclical stocks* are shares in companies whose performance is closely linked to the strength of the economy, such as firms in energy, industrials, or consumer discretionary sectors. *Defensive stocks*, by contrast, are shares in companies that tend to be less sensitive to economic downturns, such as utilities or consumer staples. This difference in sensitivity entails that defensive stocks are less risky; however, they typically underperform during economic expansions when investors favor cyclical sectors to capture rising market momentum (de Longis et al., 2022).

Investors often turn to defensive stocks for protection when an economic slump hits, then rotate back toward cyclical stocks at the first signs of recovery (Stangl et al., 2007). This shifting balance is central to the broader problem of managing portfolio risk across the business cycle. The objective is not simply to choose the best-performing sectors at a single point in time, but to build a portfolio that optimizes risk-adjusted returns across the business cycle. The cyclical-defensive distinction is thus used as a financially relevant partition through which cross-sector dependence can be studied.

Nonetheless, the usefulness of this distinction depends on how these groups behave in relation to one another. If they fall together during periods of stress, then the protection offered by the defensive part of the portfolio may be weaker than expected. For this reason, portfolio risk depends not only on the individual investments selected, or on the amount of money assigned to each one, but also on the relationships between them. These relationships can be understood as patterns of co-movement, referring tothe extent to which asset returns tend to rise or fall together. A portfolio may appear diversified because it contains many stocks or several sectors, but if those investments are all influenced by the same economic forces, then the portfolio may still be exposed to a common source of risk.

Consider, for example, two portfolios that both allocate 60% of their value to cyclical stocks and 40% to defensive stocks. At first glance, they appear equally diversified because their allocations are identical. Yet their risk may be very different. In one portfolio, the cyclical and defensive stocks may move largely independently, with a 20% correlation, so the defensive portion can help offset losses during downturns. In the other, the two groups may have a co-movement closer to 60%, meaning that losses in one part of the portfolio are likely to be accompanied by losses in the other. The difference between these portfolios is therefore not visible from the allocation alone; it lies in the pattern of relationships between the assets.

The economic forces that drive this co-movement can include changes in interest rates, inflation, investor confidence, or market-wide uncertainty, which affect many sectors at once. Hence, the relationship between cyclical and defensive stocks is not fixed but may change as market conditions evolve. For investors using sector rotation, this makes the central question not only when to move between sectors, but which patterns of *joint movement* should guide that decision.

This motivates the need for methods that can study relationships between groups of stocks rather than individual stocks alone. Canonical Correlation Analysis (CCA) provides a natural framework for this problem because it measures dependence between two multivariate panels. In simple terms, CCA asks whether two groups of variables share common patterns of movement by extracting linear combinations within each group that maximize their cross-correlation. In this thesis, the two groups may be understood as teh different sets of stocks, cyclical and defensive.

However, applying this idea to financial data is not straightforward. Stock-market data often contain many assets observed over a limited time span. For example, a dataset based on the S&P 500 includes hundreds of stocks, but only a finite number of daily, weekly, or monthly observations. This creates a problem because, with many assets, the number of possible relationships grows quickly and the data may not contain enough information to measure all of them reliably. The estimated relationships would partly reflect estimation noise rather than genuine economic relationships.

Recent theoretical work has formalised this difficulty. Bykhovskaya and Gorin (2023) study high-dimensional CCA (HDCCA), where the number of variables in the two datasets grows together with the number of observations. They show that, in this setting, classical CCA results in biased estimates. Additionally, they derive ways to assess the size of the estimation error, suggesting that the behaviour of CCA in large datasets can still be informative if treated carefully. Rather than using CCA as a perfect map of the market, the method can be used as a diagnostic tool. It can help identify whether the relationship between groups of stocks appears to be concentrated around a few common patterns (*low-rank*), or whether it is more dispersed across many different sources of movement (*diffuse*). This can be described in terms of the *spectral structure*, meaning how strongly the relationship is concentrated in a few dominant patterns, and the *geometric structure*, meaning how closely the directions of movement between the two groups are aligned.

Applied to diversification, if the relationship between cyclical and defensive stocks is largely explained by a small number of common patterns, then the portfolio may be more exposed to shared economic forces than its sector allocation suggests. If the relationship is more spread out, then the groups may provide more distinct sources of return and risk. Understanding this structure therefore helps assess whether sector rotation and diversification strategies are genuinely reducing risk, or whether they are simply rearranging investments that remain driven by the same underlying forces.

Despite these advances, it remains unknown whether these high-dimensional phenomena meaningfully describe the structure of *real* equity markets. The present study seeks to address such gap by answering the question: to what extent does HDCCA theory accurately describe the spectral and geometric structure of cross-sector return dependence in large equity panels?

This contribution is important for three reasons. First, it provides empirical evidence of HDCCA theory in real financial markets. Second, it clarifies whether canonical methods remain informative in large equity universes despite theoretical inconsistency. Third, it offers a structural perspective on cross-sector dependence that bridges random matrix theory and applied asset pricing, thereby informing both risk management and factor modelling practice.

To answer this question, we combine simulation evidence with an empirical application. The theoretical framework is reviewed in chapter 2, covering classical CCA, its breakdown in high-dimensional settings, and the Bykhovskaya-Gorin framework. Building on this, chapter 3 presents the Monte Carlo simulations, where the relevance of the HDCCA framework is assessed under controlled data-generating processes calibrated to realistic financial dimensions. Full explanations of the data-generating processes and further results can be found in Appendix A. The simulations act as a stepping stone to chapter 4, providing the intuition behind the benchmarks and the diagnostic framework in the empirical application. The results of the empirical application to large equity panels are reported in chapter 5. Finally, chapter 6 concludes with a discussion of implications and final remarks.

2. LITERATURE REVIEW

2.1. Classical Canonical Correlation Analysis

Conceptual Overview. Canonical Correlation Analysis (CCA), first introduced by Hotelling (1936), is a classical multivariate statistical method designed to measure the strength of linear association between two random vectors. Let $X_t \in \mathbb{R}^p$ and $Y_t \in \mathbb{R}^q$ denote two zero-mean random vectors observed over $t = 1, \ldots, n$. In the context of this thesis, these vectors represent panels of cyclical and defensive stock returns, respectively.

CCA seeks linear combinations of vectors $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$ that maximize the correlation between linear projections of the two variable sets:

$$\rho = \max_{a,b} corr(a^\top X, b^\top Y)$$

Define $U = a^\top X, V = b^\top Y$, which are referred to as the *canonical variables*. The vectors a and b are known as *canonical directions*, and the resulting correlation (ρ) between U and V is the *canonical correlation*.

Intuitively, CCA identifies the directions in the two variable spaces that exhibit the strongest shared variation. Formally, this corresponds to the first pair of canonical directions (a_1, b_1) which solves the optimization problem

$$(a_1, b_1) = \arg \max_{a \in \mathbb{R}^p,\, b \in \mathbb{R}^q} \frac{a^\top \Sigma_{XY} b}{\sqrt{a^\top \Sigma_{XX} a}\sqrt{b^\top \Sigma_{YY} b}}, \tag{2.1}$$

where

$$\Sigma_{XX} = \mathbb{E}[X_t X_t^\top], \quad \Sigma_{YY} = \mathbb{E}[Y_t Y_t^\top], \quad \Sigma_{XY} = \mathbb{E}[X_t Y_t^\top]$$

denote the covariance matrices within and between the two variable sets. The maximum value of this objective function is the first canonical correlation, denoted ρ_1. The first pair of canonical variables represents the strongest possible linear relationship between the two datasets.

After the first pair of canonical variables is obtained, subsequent pairs are obtained recursively under orthogonality constraints,

$$a_k^\top \Sigma_{XX} a_j = 0, \quad b_k^\top \Sigma_{YY} b_j = 0, \quad \text{for } j < k. \tag{2.2}$$

ensuring that each successive pair captures a new orthogonal mode of cross-sectional dependence between the two datasets.

Eigenvalue Representation. The result to the optimization problem defined in Equation [2.1] can be expressed through the singular value decomposition of the whitened cross-covariance matrix. Specifically, the weights are determined by the eigenvectors of the product of the inverse covariance matrices and their cross-covariances (a process known as whitening).

$$M = \Sigma_{XX}^{-1/2}\Sigma_{XY}\Sigma_{YY}^{-1/2}.$$

Since

$$C = MM^\top,$$

the eigenvalues of C are the squared singular values of M. Thus, the canonical correlations ρ_k are the singular values of M, and the canonical directions are obtained by transforming the corresponding singular vectors back to the original coordinate systems.

Equivalently, Johnson and Wichern, 2007, pp. 539–574 demonstrate in Chapter 10, how solving this constrained maximization through the method of Lagrange multipliers is equivalent to solving a generalized eigenvalue problem. CCA can then be expressed as an ordinary symmetric eigenvalue problem for the matrix

$$C = \Sigma_{XX}^{-1/2}\Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}\Sigma_{XX}^{-1/2}. \tag{2.3}$$

The eigenvalues of this matrix C correspond to the squared canonical correlations ρ_k^2, and the canonical directions for X_t correspond to its eigenvectors. Let u_k denote the eigenvector associated with the k-th largest eigenvalue ρ_k^2 (such that $Cu_k = \rho_k^2 u_k$.) Then the canonical directions for X_t are obtained as $a_k = \Sigma_{XX}^{-1/2}u_k$ and the canonical directions for Y_t as $b_k = \Sigma_{YY}^{-1/2}v_k$.

In empirical applications, population covariance matrices are unknown and replaced by their sample counterparts:

$$S_{XX} = \frac{1}{n}\sum_{t=1}^{n} X_t X_t^\top, \tag{2.4}$$

$$S_{YY} = \frac{1}{n}\sum_{t=1}^{n} Y_t Y_t^\top, \tag{2.5}$$

$$S_{XY} = \frac{1}{n}\sum_{t=1}^{n} X_t Y_t^\top. \tag{2.6}$$

The squared sample canonical correlations are thus the eigenvalues of $\widehat{C} = S_{XX}^{-1/2}S_{XY}S_{YY}^{-1}S_{YX}S_{XX}^{-1/2}$.

This matrix formulation provides the basis for the statistical analysis of CCA estimators. Since the sample canonical correlations and directions are obtained from the eigenvalues and eigenvectors of $\widehat{C}$, large-sample behavior can be studied through the convergence of the sample covariance matrices and the corresponding changes in the eigensystem.

Statistical Properties. Under the classical asymptotic framework in which p and q are fixed while $n \to \infty$, CCA enjoys the following well-established statistical guarantees:

- Sample covariance matrices (S_{XX}, S_{YY}, S_{XY}) are consistent estimators of their population counterparts.
- Sample canonical correlations $\hat{\rho}_k$ converge in probability to the population counterpart (ρ_k).
- Sample canonical directions converge (up to sign) to their population directions.

Under multivariate normality, likelihood-based inference is possible, and asymptotic distributions for canonical correlations can be derived. In this context, CCA provides a consistent and efficient method for detecting and estimating cross-dependence between two multivariate systems.

Practical Implications and Limitations. Despite these theoretical guarantees, CCA has historically received limited attention in empirical finance applications. One reason is computational and statistical fragility as CCA requires inversion of covariance matrices, which becomes unstable when the cross-sectional dimension is large relative to the sample size.

In recent times, the increasing availability of large cross-sectional datasets has renewed interest in multivariate techniques capable of capturing joint dependence structures. In finance, applications include studying relationships between asset returns and macroeconomic variables to extract joint factor structures and analyze cross-market dependence (Firoozye et al., 2023). CCA is useful in these contexts because it captures shared dependence between two systems without imposing a directional model.

Nonetheless, the classical consistency results rely critically on the assumption that dimensionality remains fixed as the sample size grows. When p and q increase proportionally with n, the spectral behavior of sample covariance matrices changes fundamentally, and classical asymptotic theory no longer applies. Understanding CCA in such settings requires a high-dimensional asymptotic framework.

2.2. Breakdown in High Dimensions

Eigenvalue Distortion and Spurious Correlations. As noted above, the assumption that the dimensions p and q (e.g.: number of assets in an equity universe) remain fixed as the sample size n increases is not usually satisfied in financial datasets. In such settings, the ratios

$$\frac{p}{n} \to c_1 > 0, \quad \frac{q}{n} \to c_2 > 0$$

are non-negligible.

When p or q is comparable to n, sample covariance matrices become unstable. If $p \geq n$, the matrix S_{XX} is singular and cannot be inverted, rendering classical CCA undefined without regularization. Even when $p < n$ but p/n is not small, eigenvalues of sample covariance matrices exhibit systematic distortion relative to their population counterparts.

Random matrix theory provides precise predictions for these distortions. Theory thus suggests that in high-dimensional settings, even when the true cross-covariance Σ_{XY} is zero, sample canonical correlations will converge to a non-degenerate limiting law, implying that apparent cross-panel dependence may arise purely from dimensionality-induced noise.

Additionally, when a finite number of population canonical correlations are present, the empirical canonical correlations follow a structured bulk–outlier pattern instead of converging to their population values. A large fraction of empirical canonical correlations accumulate within a deterministic interval (the "bulk"), while only sufficiently strong population correlations generate isolated eigenvalues outside this region.

The High-Dimensional CCA Framework. Such phenomena are formalized in *High-Dimensional Canonical Correlation Analysis* by Bykhovskaya and Gorin (2023). They consider a setting where

$$p, q, n \to \infty \quad \text{with} \quad \frac{p}{n} \to c_1 \in (0, 1), \quad \frac{q}{n} \to c_2 \in (0, 1).$$

Under suitable regularity conditions, they show that:

- The empirical distribution of squared canonical correlations converges to a deterministic limit characterized by a Wachter-type distribution.
- Population canonical correlations below a critical threshold are not individually identifiable: their sample counterparts are absorbed into the bulk.
- Only population correlations exceeding a phase transition threshold generate isolated empirical eigenvalues.

The existence of this critical threshold separating two phases (the bulk, where weak signals are absorbed into the noise, and the outliers) is called a *phase transition* phenomenon. It is analogous to the Baik–Ben Arous–Péché (BBP) phenomenon in random matrix theory as it implies that detectability of cross-panel dependence depends not only on signal strength but also on dimensional ratios c_1 and c_2.

As estimators of canonical directions are inconsistent in high dimensions, even when a population canonical correlation is strong enough to exceed the phase transition threshold and produce an isolated empirical eigenvalue, the associated sample canonical vector does not converge to its population counterpart.

Instead, the sample direction remains only partially aligned with the true direction. The limiting absolute cosine of the angle between these vectors is strictly less than one and depends explicitly on signal strength and dimensional ratios. Thus, estimation error does not vanish asymptotically; the sample direction remains separated from the population direction by a non-zero limiting angle.

Correction and Benchmark Implications. Importantly, the theory not only diagnoses high-dimensional inconsistencies but also quantifies it. Given observed empirical canonical correlations and known dimensional ratios (c_1, c_2), the framework provides formulas that map sample quantities back to the corresponding population canonical correlations. First, an isolated sample canonical correlation can be mapped back to its population canonical correlation. Second, the alignment between the sample and population canonical directions can be measured through the limiting cosine of the angle between them.

These correction formulas provide a way to interpret canonical components despite their formal inconsistency. In light of this, the Bykhovskaya-Gorin theory serves as the benchmark for the empirical analysis. Throughout this study it is evaluated whether the structural patterns they show for high-dimensional asymptotics are visible in real equity panels.

In this financial application, the canonical directions define *canonical variates*, that is, linear combinations of the variables in each panel. When the variables are asset returns, these variates can be interpreted as return-based factors. Moreover, a *canonical root* refers to the squared canonical correlation, $\lambda_k = \rho_k^2$. While *canonical correlations* refer to the co-movemnt or linear dependence between the two panels (cyclical and defensive stock returns).

In particular, the analysis examines whether the empirical canonical correlations exhibit a bulk-outlier structure and whether the instability of the estimated canonical directions aligns with the theoretical cone-angle behaviours seen in HDCCA. Evidence of these patterns would suggest that high-dimensional CCA provides a useful structural description of cross-sector dependence.

3. MONTE CARLO SIMULATIONS

To determine whether the bulk-outlier behaviour expected from HDCCA is large enough to affect inference, Monte Carlo simulations are performed. Through a controlled data-generating process, six different experiments are evaluated: a pure-noise scenario, a one-spike signal, a near-critical case, a multi-spike case, diffuse multi-spikes and finance-style robustness checks that introduce heavy-tailed observations and non-identity within-panel covariance structures usually found in financial panel data. Full descriptions of the data-generating process and the simulation results are provided in Appendix A.

Diagnostic Measures. Three main diagnostics are reported both in the simulations and the empirical analysis. Note that throughout the thesis, a canonical root refers to the squared canonical correlation, $\lambda_k = \rho_k^2$.

First, the largest squared sample canonical correlation (the leading sample root) is $\widehat{\lambda}_1 = \widehat{\rho}_1^2 = s_1(\widehat{M})^2$, and measures the strongest in-sample linear association between the two panels.

Second, to assess whether this relation is stable, the in-sample root is compared to its out-of-sample counterpart. A canonical relation is considered stable when it produces a large correlation both within the estimation window and in a held-out sample. Within each window, the canonical directions are estimated on a training subsample and then applied to held-out observations from both panels. Next, the out-of-sample root is computed as $\widehat{\lambda}_{1,\text{oos}} = \text{corr}^2\left(X_{\text{test}}\widehat{a}_{1,\text{train}}, Y_{\text{test}}\widehat{b}_{1,\text{train}}\right)$. A large gap between $\widehat{\lambda}_{1,\text{in}}$ and $\widehat{\lambda}_{1,\text{oos}}$ is interpreted as evidence of overfitting.

Third, the split-sample directional overlap measures the stability of the estimated canonical directions across two subsamples, using the absolute cosine similarity between the corresponding weight vectors. For the X-side direction,

$$\text{Overlap}_X = \left(\frac{\widehat{a}_1^{(1)\prime}\widehat{a}_1^{(2)}}{\|\widehat{a}_1^{(1)}\|\|\widehat{a}_1^{(2)}\|}\right)^2,$$

with an analogous definition for the Y-side direction.

Central Findings. The simulations yield four central findings as reported in Table 3.1. The first experiment, the pure-noise benchmark, shows that in high dimensions CCA can produce a large leading sample root even under exact independence. Figure 3.1 shows that the leading root is centered around 0.50, whereas the out-of-sample root and split-sample overlaps are essentially zero. This illustrates that large in-sample roots cannot be interpreted as evidence of genuine cross-panel structure unless they exceed what would be expected from high-dimensional noise alone. For this reason, the pure-noise distribution is used as a finite-sample benchmark.

Second, the one-spike and threshold experiments confirm the phase-transition logic defined in HDCCA. Figure 3.2 shows that weak population spikes do not generate a distinct sample outlier as the leading sample root remains close to the pure-noise benchmark, and the estimated canonical directions show little recovery. Moreover, the threshold zoom in Figure 3.3 shows that crossing $\rho_c \approx 0.426$ (the critical threshold) does not produce a sharp finite-sample discontinuity. Instead, the probability of detecting the signal as an outlier increases gradually around the threshold, and both vector recovery and held-out canonical correlation improve more slowly than the in-sample root. Taken together, the results indicate that spectral separation alone is not sufficient evidence of stable or economically meaningful cross-panel dependence.

Third, the multi-spike and diffuse designs demonstrate that detecting the signals also depends on how they are distributed. In the multiple-spike design, only the strongest components separate most clearly, while weaker components are harder to distinguish from the bulk. Annex Figures A.8–A.9 portray this point as concentrated dependence produces clearer leading roots and greater directional stability, whereas diffuse dependence produces a smoother spectrum that remains closer to the high-dimensional bulk.

Fourth, when applying the Bykhovskaya-Gorin (BG) correction formulas, the results validate the theoretical expectation under clean separated-spike designs as clearly depicted in Figure 3.5. While in the one-spike experiment shown in Figure 3.4, raw roots are severely inflated with $\rho^2 = 0.4774$ at $\rho = 0.6909$, while the mean raw root is 0.6509. In contrast, the BG-corrected estimate is 0.4754, almost unbiased. However, near or below the threshold, corrected values can still be upward biased. Thus, correction is reliable only after separation.

Note that the robustness exercise in Figure 3.7 shows that the main conclusions survive moderate covariance misspecification. Detection remains strong when the clean Gaussian benchmark is relaxed and introducing within-panel covariance has limited effects in this calibration, whereas heavy-tailed observations do increase the in-sample root and weaken directional recovery and stability. The same occurs when the BG correction is applied, as the corrected root remains close to the true population value for Gaussian data but remains upward biased under heavy tails.

These results motivate the empirical methodology developed in Chapter 4. The simulations show that a large leading sample root is informative only when it is large relative to an appropriate high-dimensional or finite-sample benchmark and is supported by stable canonical directions and held-out performance. Subsequently, the presence of several nonzero roots is not, by itself, evidence of a strongly identified low-rank dependence structure, since diffuse weak links can generate a broad spectrum without producing robust canonical directions. The empirical analysis therefore evaluates the cyclical and defensive panels using a joint set of diagnostics: spectral magnitude, benchmark comparison, eigengap separation, vector and subspace stability, and out-of-sample canonical correlation. Additionally, the simulations show that BG corrections are informative only when the empirical root is clearly separated from the high-dimensional bulk. This approach avoids interpreting in-sample fit alone as evidence of economically meaningful co-movement.

4. METHODOLOGY

4.1. Objective and empirical strategy

Guided by the Monte Carlo evidence, the following empirical analysis is designed to evaluate whether HDCCA provides a useful description of the dependence between cyclical and defensive stock returns. To do this, classical CCA is applied to rolling windows of stock returns and the results are then evaluated using the high-dimensional and finite-sample benchmarks developed above. The analysis is centred around three main questions. First, are the leading canonical roots larger than would be expected under high-dimensional noise? Second, are the estimated canonical directions stable across samples? Third, do they remain informative out of sample?

Through these diagnostics, the purpose is to evaluate whether the high-dimensional effects characterized by Bykhovskaya and Gorin are visible in the data, particularly inflated in-sample roots, bulk-outlier separation, and instability of the estimated canonical directions. As shown in Appendix A, in high dimensions, large sample roots do not necessarily imply that the corresponding canonical vectors are reliably estimated. For that reason, the empirical analysis does not rely on in-sample fit alone and instead evaluates spectral strength, stability, and out-of-sample performance jointly.

4.2. Empirical setting

The empirical application uses weekly stock returns constructed from CRSP data. The stocks are divided into two balanced panels, cyclical and defensive, based on FF12 industry groups derived from SIC codes. This partition is economically motivated by the business-cycle distinction and is related to Bykhovskaya and Gorin's own empirical illustration. Cyclical stocks are defined as firms in durable goods, manufacturing, energy, chemicals, and retail or services, while defensive stocks are defined as firms in non-durables, health, telecom, and utilities. Further data details and sample composition are reported in Chapter 5.

4.3. Empirical diagnostics

Empirical results are interpreted using a joint diagnostic criterion: Evidence for a strong and economically meaningful low-rank structure requires three conditions at the same time: the leading root must lie clearly above the bulk and pure-noise benchmarks, the associated canonical directions must be reproducible across subsamples, and the estimated dependence must persist out of sample. Otherwise, a large in-sample canonical root is not treated as evidence of a reliably recoverable canonical direction. To evaluate these conditions, six diagnostics are computed for each window.

First, in each window, the observed canonical roots are compared with the HDCCA bulk edge λ_+ and with a permutation-based null to determine whether the leading root is larger than would be expected under high-dimensional noise. The permutation null refers to the distribution of the leading root obtained by randomly permuting the time ordering of one panel relative to the other, which breaks any genuine cross-panel dependence while preserving the marginal structure of each panel. Additionally, the number of canonical roots above the bulk edge in each window is also used as a diagnostic. This second diagnostic helps distinguish a single dominant factor from broader multi-direction dependence.

Roots above this threshold provide evidence of cross-panel dependence relative to the specified high-dimensional benchmark; however, the corresponding vectors must also be stable. To measure such directional stability each window is split into two halves and the leading canonical vectors across the two subsamples are compared. Low overlap would indicate that the leading canonical portfolio is unstable.

This third measure is complemented by top-k subspace stability, which evaluates the overlap of the leading k-dimensional subspaces across subsamples. This is important because even when the first vector is unstable, the leading subspace may still be well-defined if there are several closely spaced roots. In other words, instability of the first vector may reflect either genuine noise or simple rotation within a weakly separated leading subspace.

The fifth diagnostic is eigengap separation: $\lambda_1 - \lambda_2$, $\lambda_2 - \lambda_3$, and $\lambda_3 - \lambda_4$. These show whether the leading root is sufficiently separated to be interpreted as an individual component. This step is essential as, when eigengaps are too small, BG inversion formulas cannot be applied credibly.

Finally, out-of-sample performance is evaluated: canonical weights estimated in one part of the data are applied to the other part, and the held-out leading squared canonical correlation is recorded. This shows whether the apparent dependence survives out of sample.

By relying on these six diagnostics jointly, the empirical analysis can evaluate whether large in-sample canonical roots are accompanied by stable and economically meaningful canonical directions, or whether they are more consistent with high-dimensional noise and instability.

4.4. Rolling-window design

The baseline specification uses rolling windows of a fixed length of 260 weeks and fixed step size of 4 weeks, yielding 144 windows over the sample. This design allows the canonical spectrum to vary over time while keeping the empirical setting explicitly high-dimensional. For each window, the six diagnostics are computed and a leading root is treated as credibly identified only if the first root is clearly separated from the second one.

4.5. Robustness

The main results are checked using alternative window lengths and subsamples. Shorter and longer windows test sensitivity to the time-series dimension, while pre-2020 and post-2020 subsamples test whether the results depend on a particular market period.

5. EMPIRICAL APPLICATION TO STOCK RETURN PANELS

5.1. Data description

The HDCCA framework is applied to a large weekly equity panel constructed from CRSP daily stock data. The sample contains 832 weekly observations from 2010 to 2025, with two balanced cross-sections of 80 cyclical and 80 defensive stocks. Notably, the cyclical group is composed mainly of retail/services, manufacturing, energy, and chemicals, while the defensive group is concentrated in health, utilities, non-durables, and telecom. This ensures that the two panels represent distinct sectors of the economy and differ in their sensitivity to business-cycle conditions.

5.2. Main result

Table 5.1 summarizes the baseline 260-week rolling specification. These windows are stepped every four weeks, resulting in 144 rolling windows. The settings are therefore high-dimensional, with p=q=80 and n=260, allowing the use of HDCCA.

The leading empirical root is large, with a mean value of 0.9731 across windows, compared with a mean HDCCA bulk edge of 0.8521. The leading root exceeds the bulk edge in every window, and the mean number of roots above the bulk is 4.17. This is supported by Figure 5.1 where the leading root is consistently too large to be explained either by the HDCCA bulk edge or by high-dimensional noise alone, as represented by the permutation null. Figure 5.2 further shows that the evidence is not confined to a single isolated component as multiple roots lie above the bulk in all windows. This suggests that the cross-panel dependence is multidimensional at the spectral level, rather than being driven only by the first canonical relation. Overall, the effect is strong enough to suggest that, at the level of the sample spectrum, cyclical and defensive returns exhibit persistent and meaningful cross-panel dependence.

The full-sample canonical score plots in Figure B.2 provide complementary visualization. The first score pair displays a much tighter linear relationship, while the second and third pairs are still positively related but weaker and more diffuse.

However, the estimated first canonical direction is not reproducible across halves of the sample. The mean split-sample squared cosine overlap of the first vector is only 0.0219 on the cyclical side and 0.0329 on the defensive side, as reported in Table 5.1 and Figure 5.3. These values are extremely low and imply that the first canonical directions extracted in one half of the data contain almost no directional information about the first canonical portfolio extracted in the other half.

The same instability appears out of sample as the mean held-out leading squared root is only 0.0128, which is very small relative to the mean in-sample value of 0.9731, as depicted in Figure 5.4. This is precisely the gap highlighted by the Monte Carlo analysis. A large in-sample canonical root is not enough to be economically meaningful; it must also survive out of sample.

The overall implication is that the leading empirical root is observed as a spectral phenomenon, but the leading empirical vector is not recoverable as a stable economic portfolio. In other words, the spectrum indicates detectable cross-sector dependence, but the first canonical direction does not behave like a reliable low-rank factor.

5.3. Top-k subspace stability and the rotation problem

The weak split-sample stability of the first vector raises an important ambiguity. A low top-1 overlap does not by itself prove that the leading dependence structure is meaningless. It may instead reflect a rotation problem: if the top few roots are close to each other, different samples may estimate different linear combinations inside essentially the same low-dimensional subspace.

To address this possibility, the analysis examines top-k subspace overlap for $k = 1, 2, 3, 5$. Figure 5.5 shows that the top-k subspaces are somewhat more stable than the top-1 vectors, suggesting that part of the first-vector instability may be due to rotation among the leading directions rather than pure noise at a single rank. Nonetheless, the gains are modest and the top-5 subspace overlap remains low, at 0.0958 on the cyclical side and 0.1297 on the defensive side (see Table 5.1).

As the leading dependence space is only weakly identified despite allowing for rotational ambiguity, the evidence is more consistent with clustered or diffuse multi-direction dependence than with a sharply separated low-rank structure.

5.4. Eigengaps and the failure of credible inversion

Eigengap diagnostics reinforce this interpretation. Figure 5.6 shows that the gaps between the leading roots are too small to support credible leading-root inversion. In the baseline run, there are zero credible leading windows out of 144, i.e.: no rolling window satisfies the separation rule required to safely apply the Bykhovskaya-Gorin outlier formulas on a window-by-window basis.

This finding has two consequences. First, it explains why the first-vector overlaps are so low: when λ_1 and λ_2 are not well separated, the estimated leading eigenvector can rotate substantially across samples even if the broader leading space contains real dependence. Second, it implies that successful empirical recovery of population signal strength from rolling outliers cannot be claimed.

Accordingly, a phase-transition recovery exercise cannot be performed credibly as the leading roots are spectrally strong but geometrically fragile. Nonetheless, for the purpose of outlier diagnostics the BG corrections are applied to the leading roots above the bulk edge. Results can be seen in Table 5.2 and Figure 5.7. This exercise shows that there are two persistent outliers that appear in 100% of windows, whereas the third signal only appears in 83%. For the two stronger signals the mean BG-corrected ρ^2 values are large, at 0.929 and 0.767 respectively. The third signal shows much weaker BG-corrected values that seem to blend with the next outliers. Thus, the data does show some evidence of persistent multi-dimensional dependence, but the lack of credible inversion and the weak directional stability suggest that this dependence is not organized into clean, well-separated canonical components.

5.5. Interpreting the spectrum

Taken together, the spectral and geometric evidence points to a strong and persistent cross-sector dependence between cyclical and defensive stock returns. It is not plausibly attributable to noise alone. But at the same time, the leading empirical canonical portfolio is not a stable object as the estimated first direction changes materially across halves of the sample and fails to transport out of sample.

Evidence suggests a diffuse structure in which several nearby directions contribute to cross-panel dependence, but no single one is stably estimated as the dominant direction. That reading also matches the Monte Carlo comparison between concentrated low-rank and diffuse dependence designs, where broad nonzero spectra can coexist with weak directional recovery.

5.6. Robustness

The robustness exercises show that the main conclusion is not driven by the baseline window length or by the specific pre-/post-2020 sample split as seen in Table 5.3 and Figure 5.8.

With shorter 208-week windows, the leading root remains an outlier in every window, but vector stability, top-5 subspace overlap, and held-out performance all remain very weak, with no credible leading windows.

With longer 364-week windows, spectral separation becomes stronger and the mean held-out root rises materially to 0.3385, well above the baseline value of 0.0128. Longer windows allow a larger part of the broad cross-panel dependence to survive out of sample, which is consistent with lower estimation noise and with more persistent low-frequency co-movement being captured over longer horizons. However, this improvement does not solve the main identification problem. First-vector overlap remains very low, top-5 overlap stays near 0.10, and the number of credible leading windows is still zero, failing to deliver stable leading canonical components that could be interpreted.

The subsample analysis leads to the same conclusion. In both the pre-2020 and post-2020 samples overlap measures remain low, held-out roots remain small, and no credible leading windows emerge. However, the patterns post-2020 point to stronger broad-spectrum dependence beyond the first two canonical ranks. This is visible in the higher outlier-count behaviour in Figure 5.2 and in the cross-specification comparison in Figure 5.8, where the post-2020 specification shows stronger evidence of dependence across several leading directions. Financially, this pattern is consistent with the idea that crisis and post-crisis market conditions compress diversification benefits, meaning that cyclical and defensive stocks become more jointly driven by common aggregate shocks so dependence appears not only in the leading canonical relation but also in weaker ranks such as ranks 3 to 5. Although the main conclusion is unchanged, this pattern can provide financial intuition.

Overall, the robustness results confirm the baseline message that the spectrum consistently indicates cross-sector dependence, but the leading canonical portfolio is not a well-identified economic direction under reasonable alternative specifications.

6. DISCUSSION AND CONCLUSION

This paper studied whether HDCCA provides a useful description of cross-sector dependence in large stock-return panels. The empirical application suggests that HDCCA is useful as a diagnostic framework, as the canonical spectrum displays strong and persistent dependence between cyclical and defensive stocks. However, the leading canonical directions are not recovered as a stable economic direction. Vector stability is low, out-of-sample performance is weak, and the leading root is not sufficiently separated from nearby roots to support credible inversion.

This distinction is central to the interpretation of classical CCA in high dimensions. A large in-sample canonical root can contain genuine information about cross-panel dependence even when the leading estimated direction is too fragile to support a clean economic interpretation.

In practice, HDCCA helps explain why large canonical roots in high-dimensional data must be interpreted cautiously. Specifically for large financial panels, classical CCA should not be interpreted using in-sample fit alone. For portfolio construction, risk management, or factor interpretation, large canonical roots are only economically meaningful if they are also stable across samples and persist out of sample. Otherwise, the estimated canonical portfolio may reflect high-dimensional estimation error rather than a reliable source of common risk.

More broadly, the results suggest that cross-sector dependence in equity markets may be real but more diffuse than a simple low-rank model would imply. The evidence is therefore more consistent with several weakly separated directions than with one clean and stable leading canonical factor.

There are several limitations to the study as the empirical analysis focuses on one panel design, one frequency, and classical CCA interpreted through the HDCCA benchmark. It does not compare alternative regularized estimators in detail, nor does it study other asset partitions or richer time-series structures. These limitations point naturally to future research. Future extensions could include a comparison of classical CCA with regularized or sparse alternatives and examine whether economically meaningful information survives more clearly at the subspace level than at the level of the first canonical vector.

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Monte Carlo Simulation Results

Design	Population signal	Spectral evidence	Validation evidence
Exp.1: independent panels	None ($\rho_j = 0$)	mean $\hat{\lambda}_1 = 0.505$ 99% cutoff=0.545	OOS=0.004 split X=0.012
Exp.2: one-spike	$\rho = 0.691$ $\rho^2 = 0.477$	mean $\hat{\lambda}_1 = 0.651$ raw bias=0.174	OOS=0.035 split X=0.053 angle X=29.3°
Exp.3: threshold zoom	$\rho_c \approx 0.426$ upper grid $\rho = 0.546$	detection=0.061 at threshold 0.598 at $\rho = 0.546$	OOS=0.004 to 0.006 angle X=68.9 to 47.2°
Exp.4: three-spike low-rank	$\rho = (0.80, 0.55, 0.30)$	mean outliers=1.589 roots=0.755/0.552/0.494	$\cos^2_X$=0.872/0.466/0.041 valid BG=1.0/0.9/0.03
Exp.5: low-rank vs. diffuse	rank 3, energy=1.243 rank 10, energy=0.762	outliers=2.108/0.130 $\hat{\lambda}_1$=0.735/0.527	OOS=0.140/0.005 split X=0.19/0.013
Exp.6: Gaussian / spiked cov. / Student-t	one spike $\rho^2 = 0.49$	detection=1.000 all roots=0.658/0.658/0.714	OOS=0.039/0.041/0.030 split X=0.061/0.063/0.041

The simulations show that large sample roots can arise under high dimensionality, even when the two panels are independent (experiment 1). Detectability increases only gradually around the phase-transition threshold (as seen in the threshold zoom; experiment 3), and vector recovery generally lags behind spectral separation. Large in-sample roots can therefore overstate the recoverable dependence. trong, well-separated low-rank components are recovered more reliably, whereas weak or diffuse dependence is harder to separate from the high-dimensional bulk (experiments 4 and 5). Under finance-style deviations from the Gaussian benchmark, heavy tails amplify in-sample roots and reduce recovery quality (experiment 6).

Table 3.1
Summary of the main Monte Carlo results.

Figure 3.1
Experiment 1 pure-noise distribution of the leading sample canonical correlation. The vertical lines report the 95th and 99th percentiles of the leading-root distribution generated.

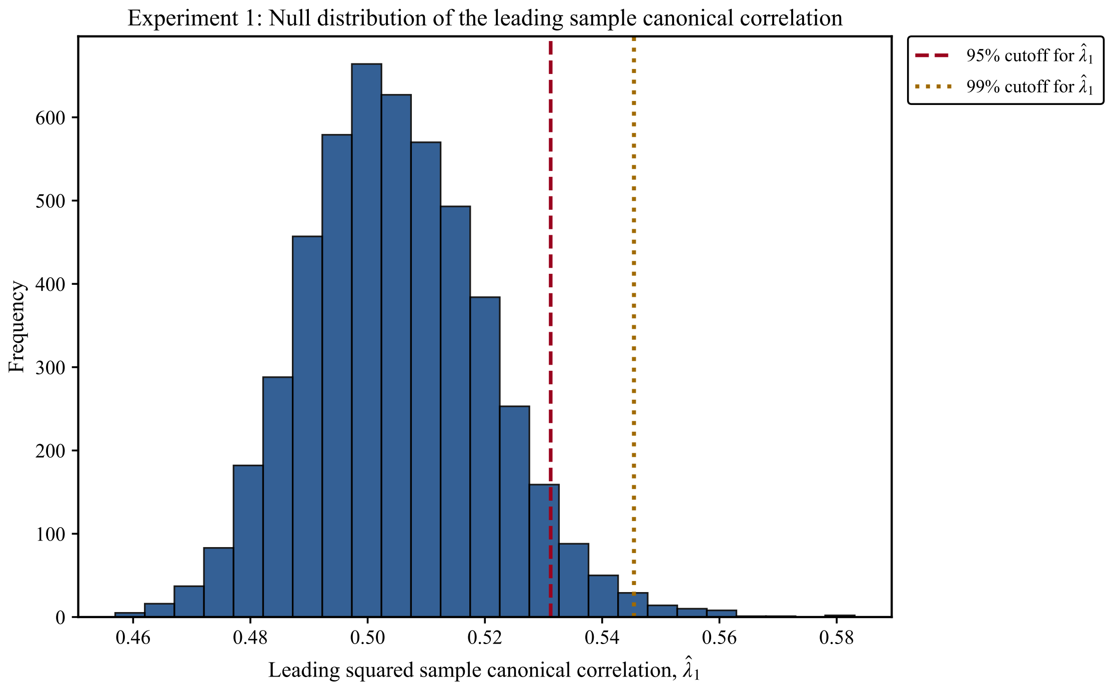


Figure 3.2
Experiment 2 in-sample spectral response to one true spike. The figure reports the mean leading and second sample canonical roots as the population canonical correlation ρ increases. The leading root remains near the high-dimensional noise level for weak signals and separates from the bulk only above the critical value ρ_c *(represented by the vertical line). The second root remains close to the bulk level, consistent with a one-spike design.*

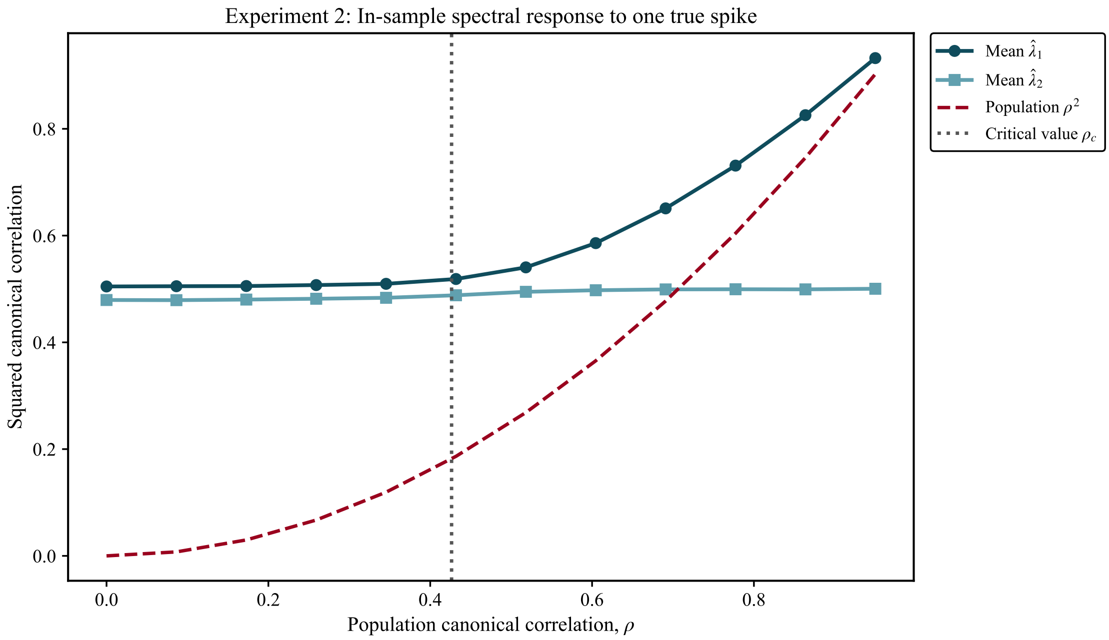


Figure 3.3
Experiment 3: Detection probability around the critical threshold. The figure reports the fraction of Monte Carlo replications in which the leading root is detected. Detection probability is low below and near ρ_c*, and increases only when the population canonical correlation moves further above the critical value.*

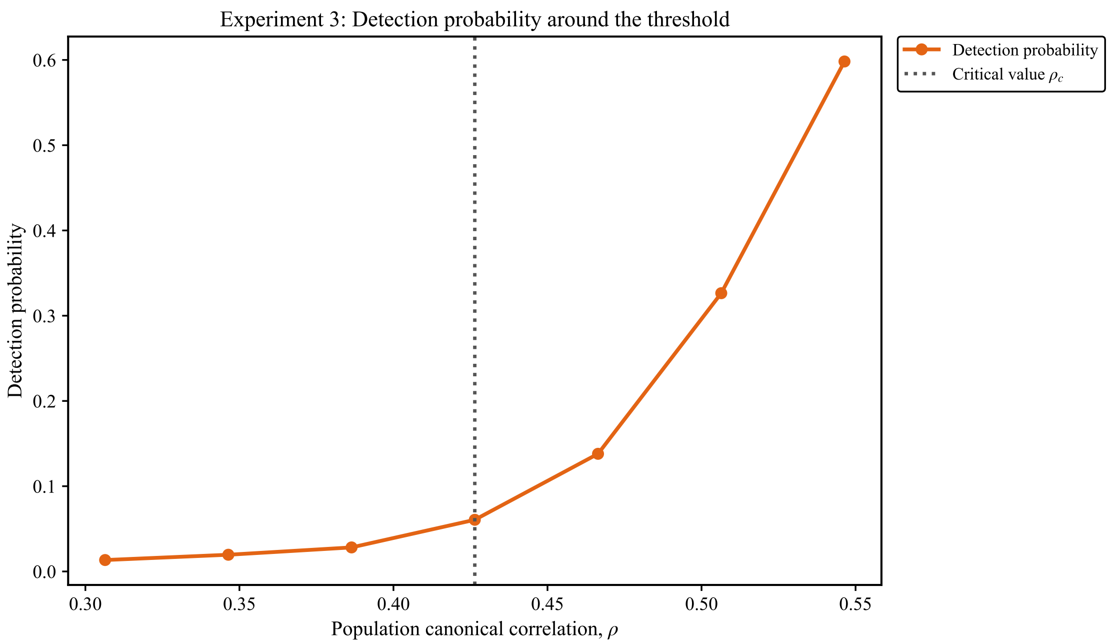


Figure 3.4

Experiment 3: BG-corrected leading root around the critical threshold. The figure reports the mean leading root before and after applying the Bykhovskaya-Gorin correction. The raw leading root is upward biased relative to the population value, especially near the threshold. In contrast, the BG-corrected root provides a much closer estimation of the population value.

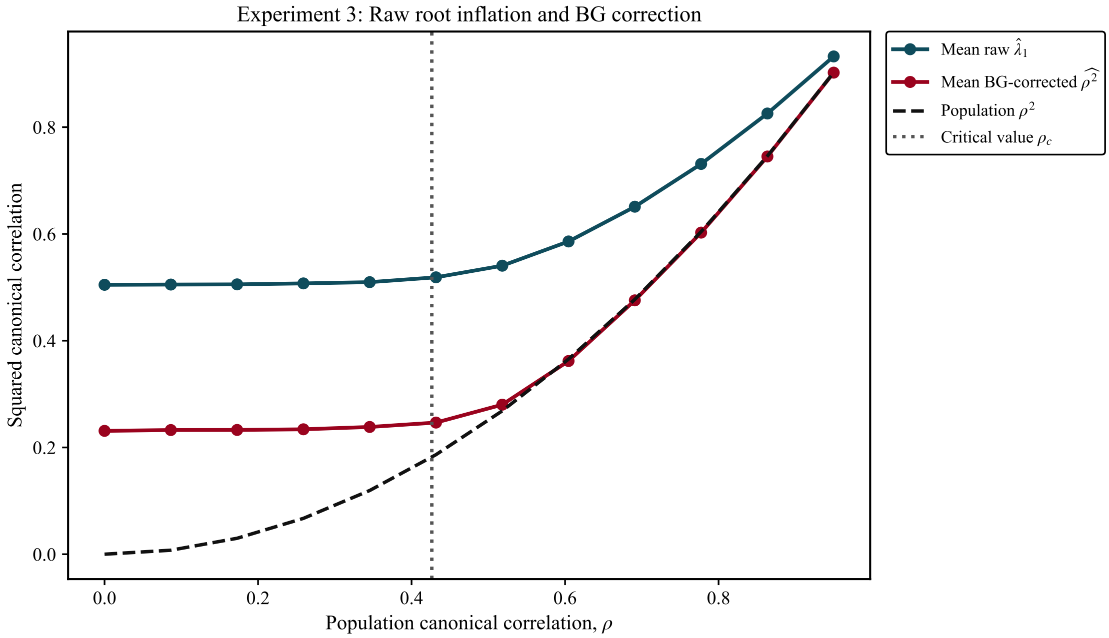


Figure 3.5

Experiment 4 BG correction by canonical component The figure reports the mean leading root before and after applying the Bykhovskaya-Gorin correction for each of the three leading spikes. Additionally, the true population value is reported for reference. It can be seen that the raw leading roots are upward biased relative to the population values. The BG-corrected roots are much closer to the population values for the first two components, while the BG-corrected root for the third spike remains inflated.

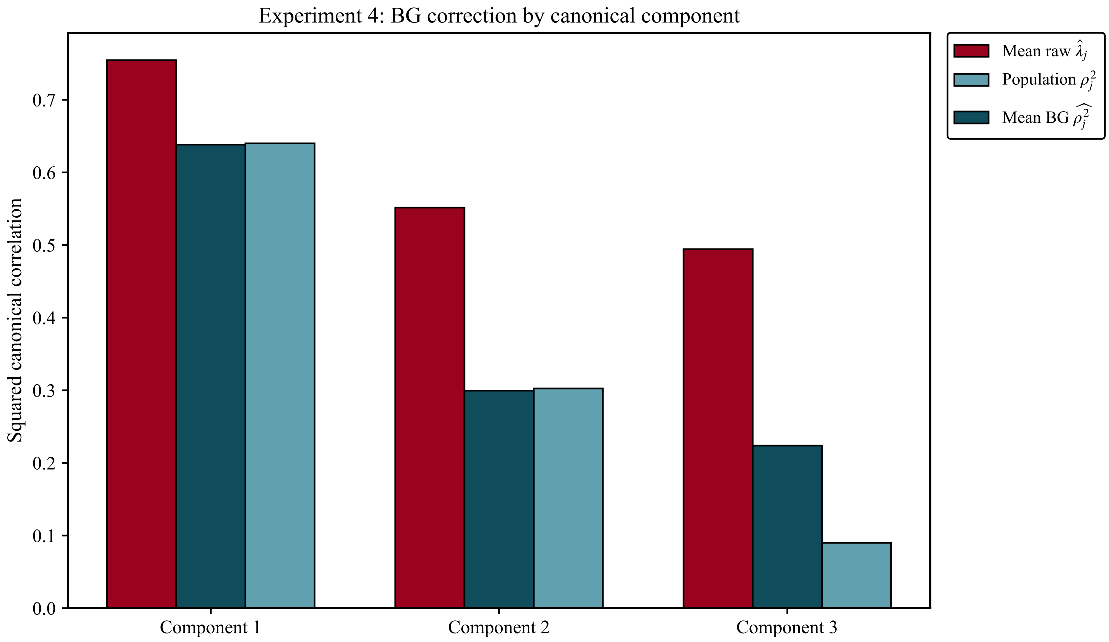


Figure 3.6

Experiment 5: BG-corrected leading root comparison between the low-rank and diffuse designs The figure reports the mean leading root before and after applying the Bykhovskaya-Gorin correction for both the low-rank and diffuse designs. The low-rank design produces a much larger raw leading root than the diffuse design, which is consistent with the fact that the signal is concentrated in a small number of directions.

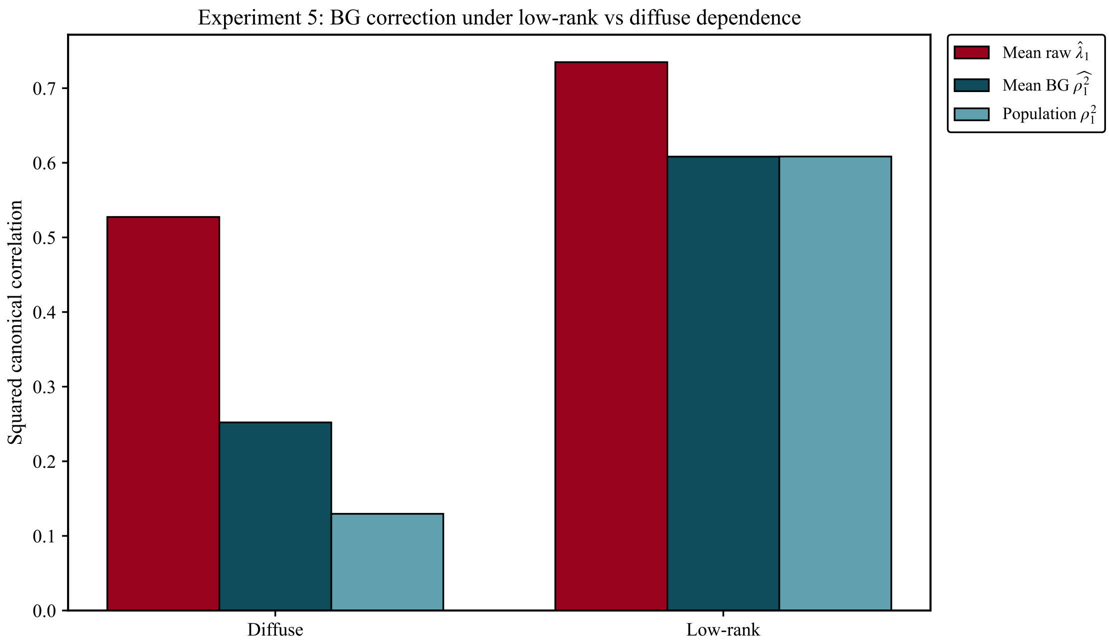
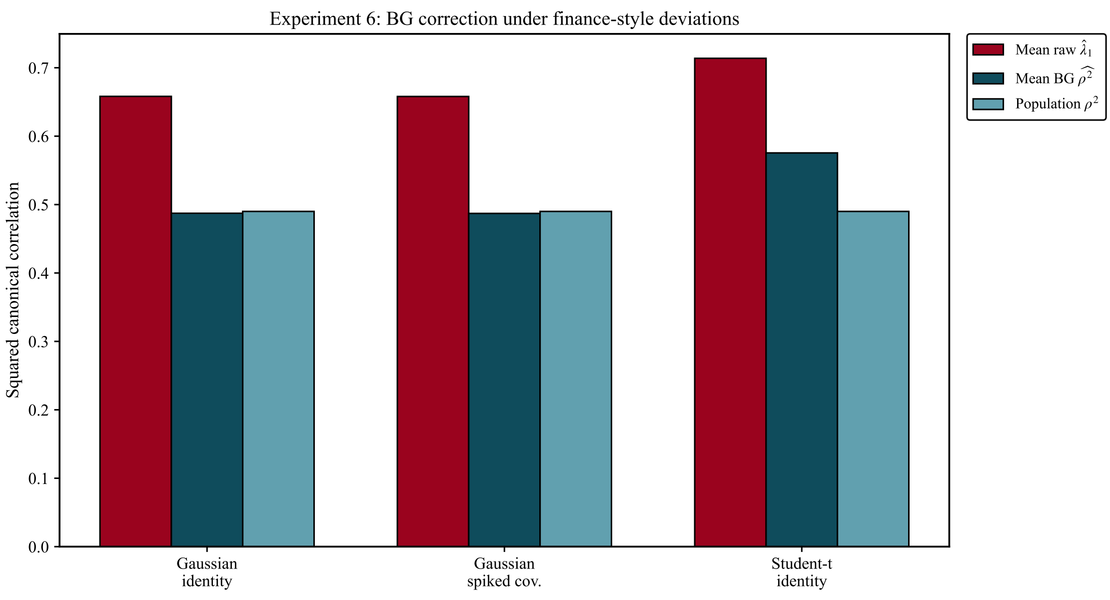


Figure 3.7

Experiment 6: BG-corrected leading root The figure reports the mean leading root before and after applying the Bykhovskaya-Gorin correction for the three cases. While BG formulae provide best results for signal recovery under Gaussian assumptions, they seem to not perform as well under non-Gaussian conditions such as t-Student.



Empirical Analysis for CRSP Stock Return Panels

Metric	Value
Rolling windows	144
Window length (weeks)	260
Panel dimensions (p, q)	$(80, 80)$
Mean leading squared root $\hat{\lambda}_1$	0.9731
Mean HDCCA bulk edge λ_+	0.8521
Share with $\hat{\lambda}_1 > \lambda_+$	100.0%
Mean number of outliers above bulk	4.17
Mean split stability, cyclical u_1	0.0219
Mean split stability, defensive v_1	0.0329
Mean held-out leading squared root	0.0128
Credible leading windows	0 / 144
Mean top-5 overlap, cyclical side	0.0958
Mean top-5 overlap, defensive side	0.1297

Table 5.1
Baseline 260-week rolling summary for the CRSP empirical application.

Rank	Outlier windows	Share	Mean $\hat{\lambda}_r$	Mean excess	Mean eigengap	Mean BG $\hat{\rho}_r^2$	Mean BG θ
1	144	1.000	0.973	0.121	0.056	0.929	11.848
2	144	1.000	0.917	0.065	0.034	0.767	25.273
3	119	0.830	0.892	0.040	0.019	0.675	34.674
4	100	0.694	0.881	0.029	0.020	0.635	38.679
5	67	0.465	0.871	0.019	0.026	0.595	43.332

Table 5.2
Rank-level summary of Bykhovskaya-Gorin diagnostics of outliers Importantly remember that none of the identified outliers pass the established credibility criterion (sufficient separation between signals).

Specification	Windows	Mean $\hat{\lambda}_1$	Mean λ_+	Mean held-out root	Mean top-5 cyclical
Baseline 260w	144	0.9731	0.8521	0.0128	0.0958
Short 208w	157	0.9839	0.9467	0.0205	0.0991
Long 364w	118	0.9612	0.6859	0.3385	0.0995
Pre-2020 260w	66	0.9657	0.8521	0.0108	0.0953
Post-2020 260w	13	0.9638	0.8521	0.0171	0.1031

Table 5.3
Robustness summary across alternative rolling-window lengths and subsamples. Across specifications, the mean leading root remains above the corresponding HDCCA bulk edge, confirming that the spectral evidence is robust to these sample choices. However, the held-out root and stability measures are much weaker and less uniform across specifications. The credible share is zero in all cases, indicating that the empirical canonical directions do not satisfy the established credibility criterion under these robustness checks.

Figure 5.1

Baseline rolling leading squared canonical root against the HDCCA bulk edge and the permutation null benchmark. The leading root remains well above both benchmarks throughout the sample, indicating persistent in-sample spectral dependence between the cyclical and defensive panels, rather than a pattern that appears only in isolated windows.

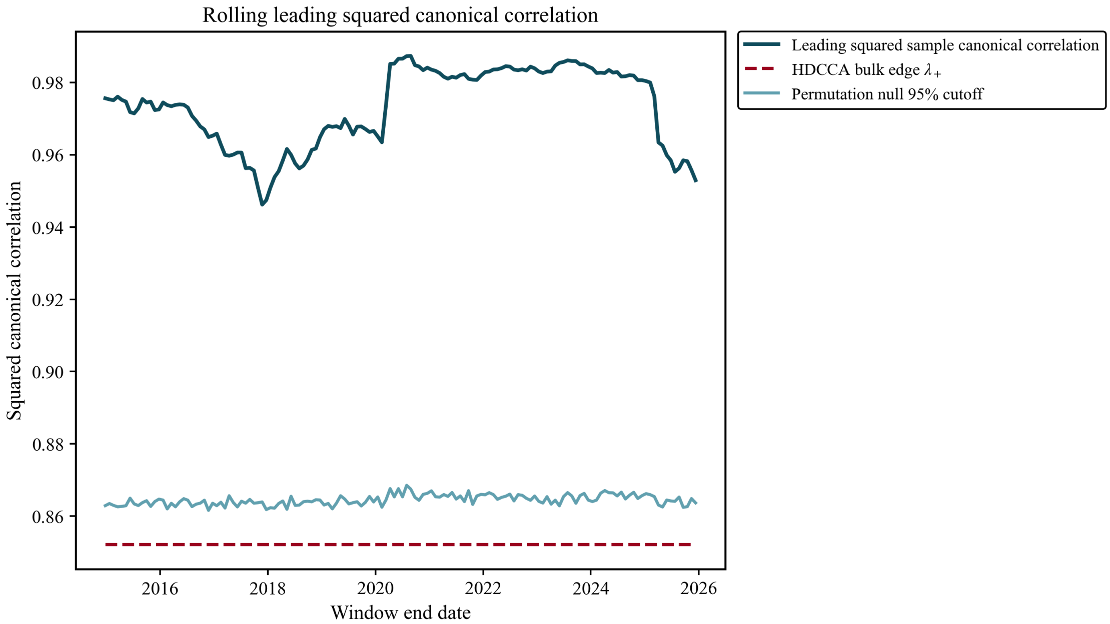


Figure 5.2

Baseline rolling count of squared canonical roots above the HDCCA bulk edge. The count is consistently positive and varies over time, with more roots exceeding the bulk edge in the middle part of the sample. This may suggest that the empirical dependence is not limited to the first canonical root.

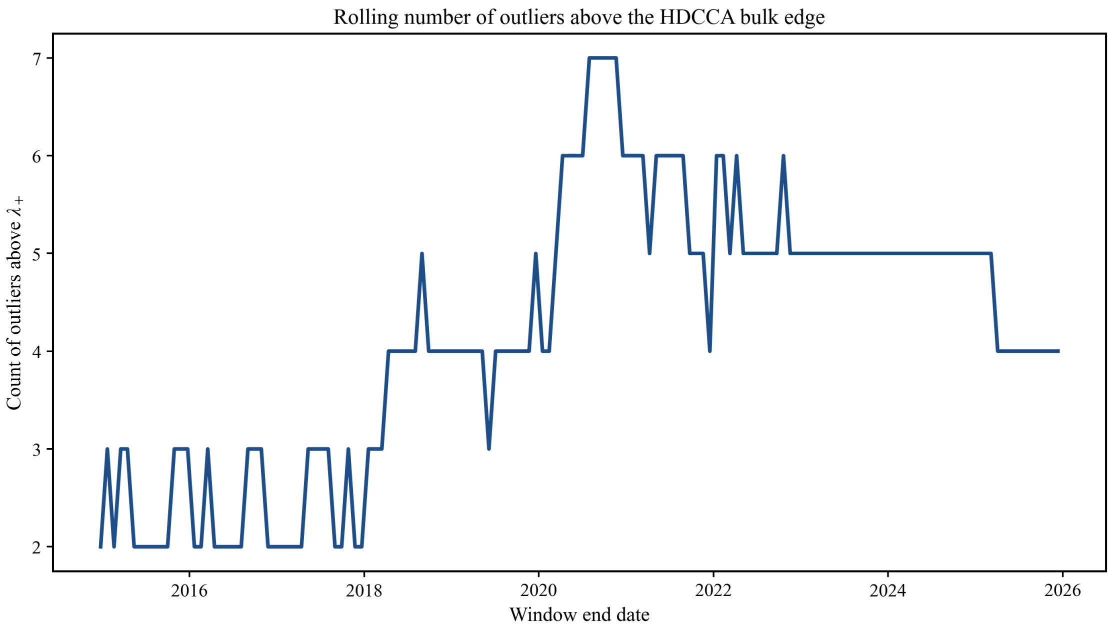


Figure 5.3

Baseline split-sample stability of the first canonical direction on the cyclical and defensive sides. The squared cosine overlaps remain close to zero for most rolling windows, implying that the leading weight vectors are not consistently recovered across subsamples, despite the large in-sample canonical root.

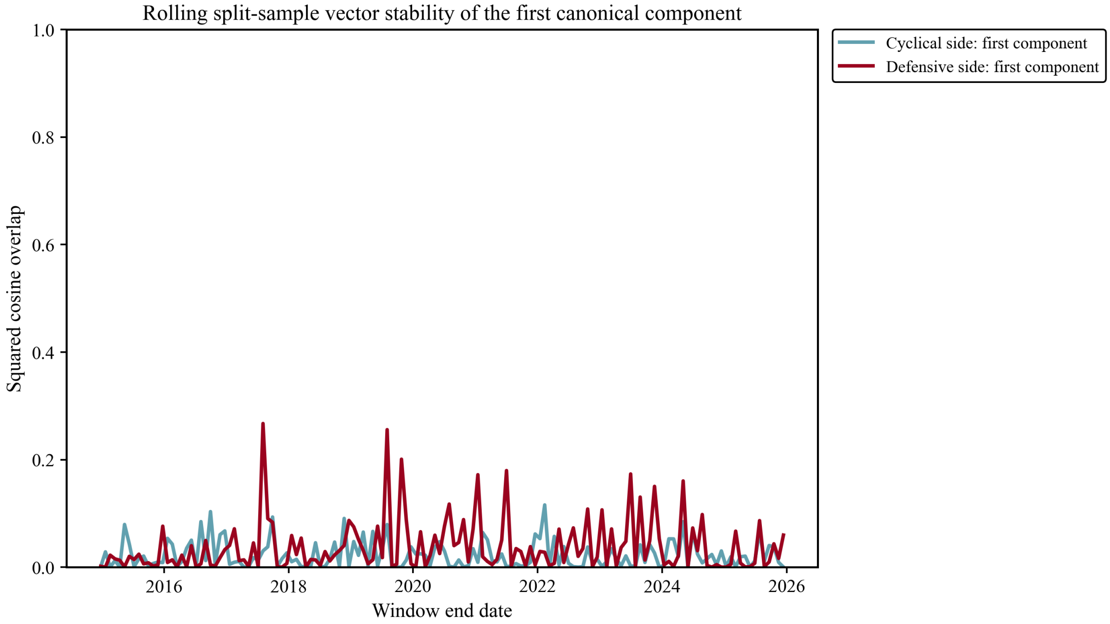


Figure 5.4

Baseline comparison between the in-sample leading squared root and the held-out leading squared root. The in-sample root is persistently large, while the held-out root remains close to zero in most windows. This large gap indicates that the strongest in-sample canonical relation does not translate into comparable out-of-sample dependence.

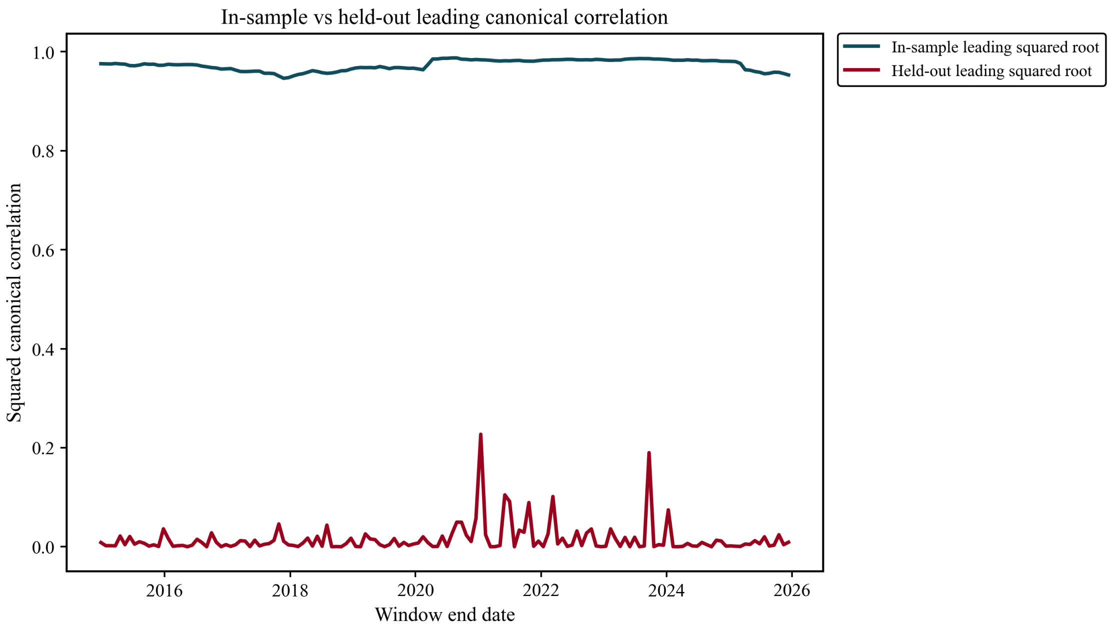


Figure 5.5

Baseline top-k subspace overlap for k = 1, 2, 3, and 5 on the cyclical and defensive sides. Allowing for several leading directions slightly increases the overlap in some windows, but the values remain low overall. Thus, the lack of stability is not only a feature of the first canonical direction; it also persists at the level of the leading canonical subspaces.

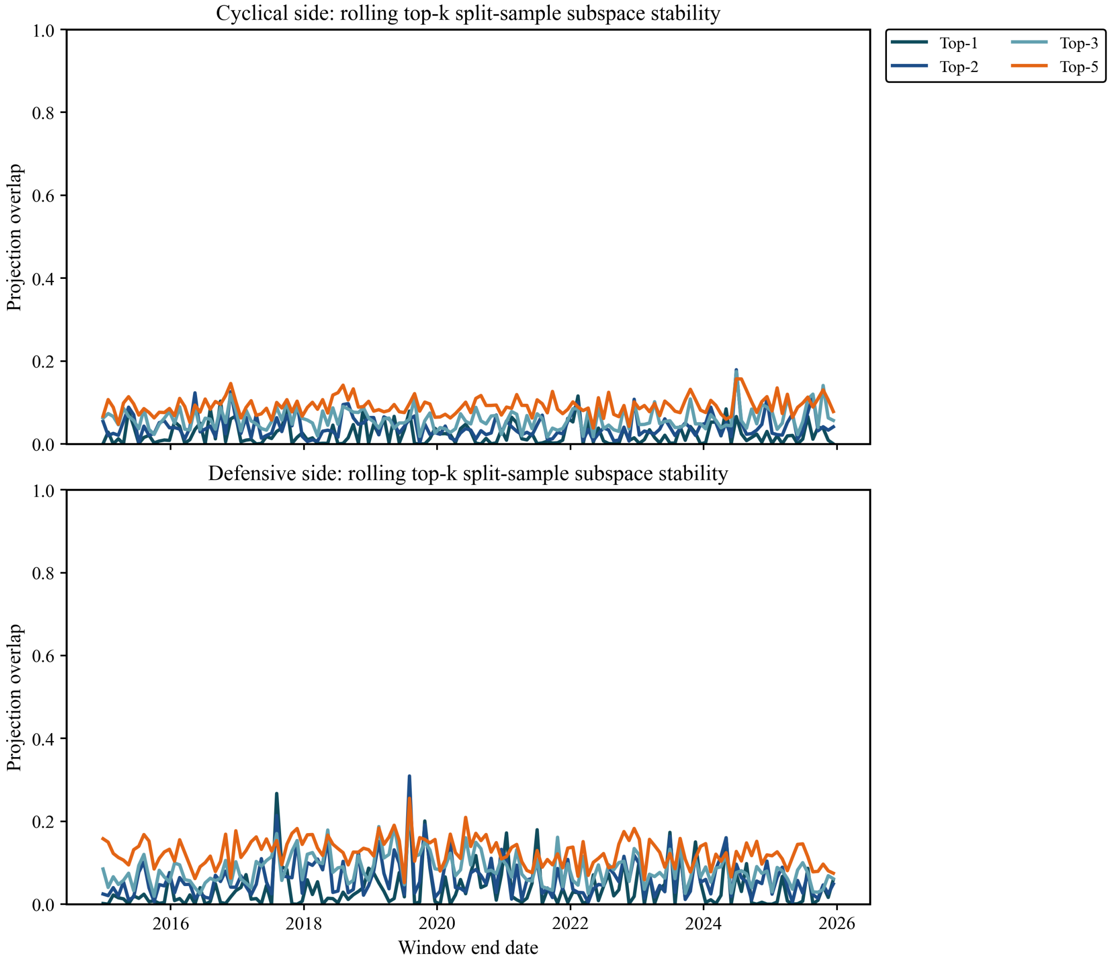


Figure 5.6

Baseline eigengap diagnostics for the first few rolling canonical roots. The eigengaps remain well below the credibility rule throughout the sample, suggesting that the leading roots are not sharply separated from neighboring roots, which makes the associated canonical directions harder to interpret as individually well-identified components.

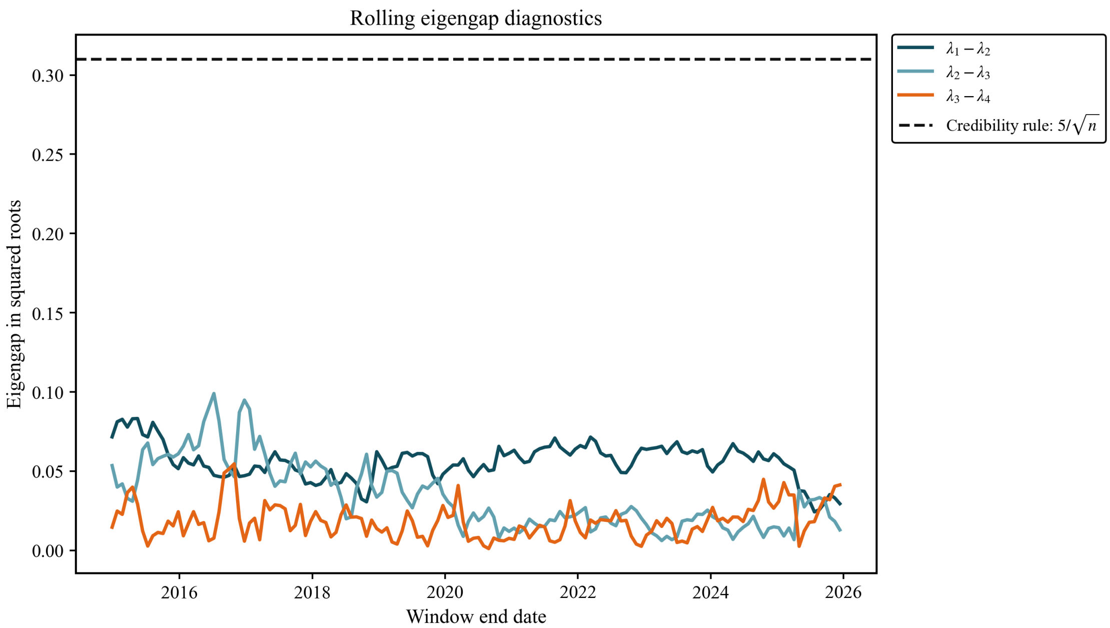


Figure 5.7

Outlier BG implied population signal strength and predicted cone-angle diagnostics.

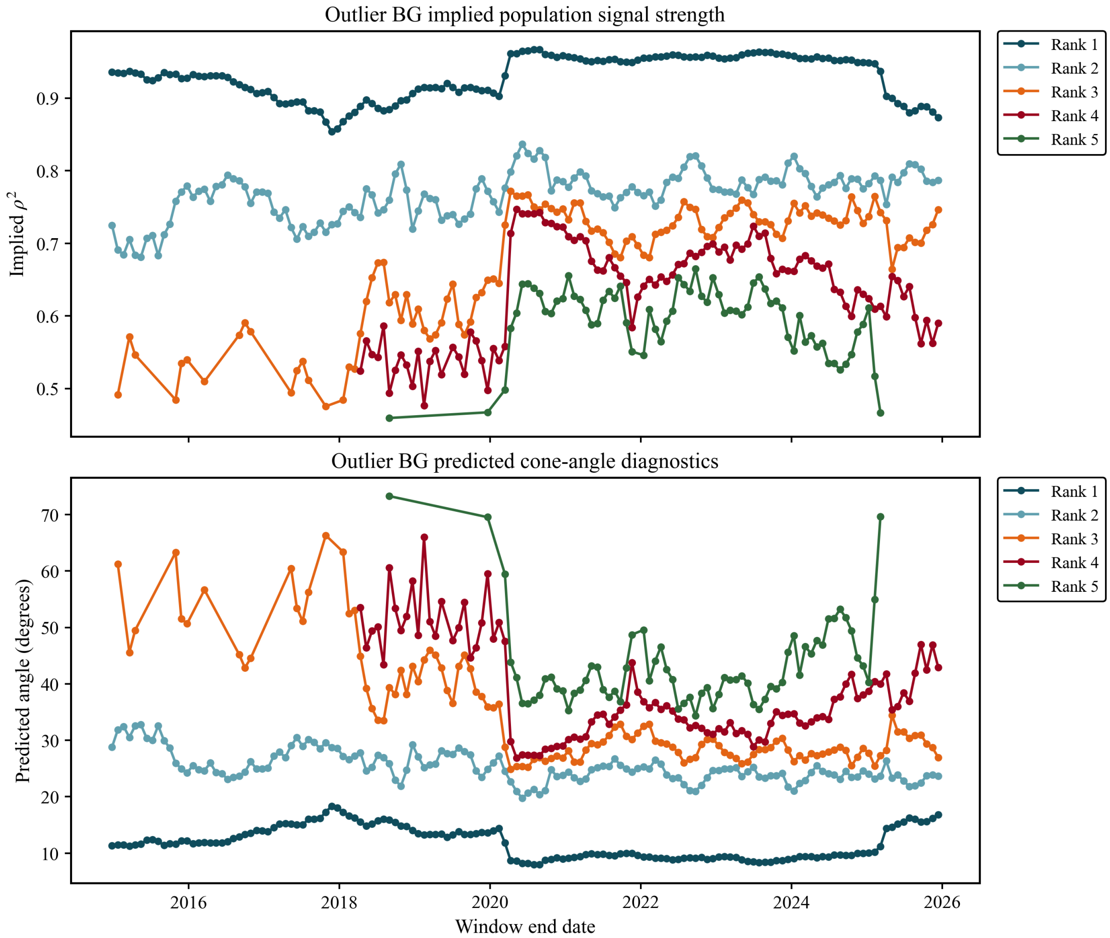
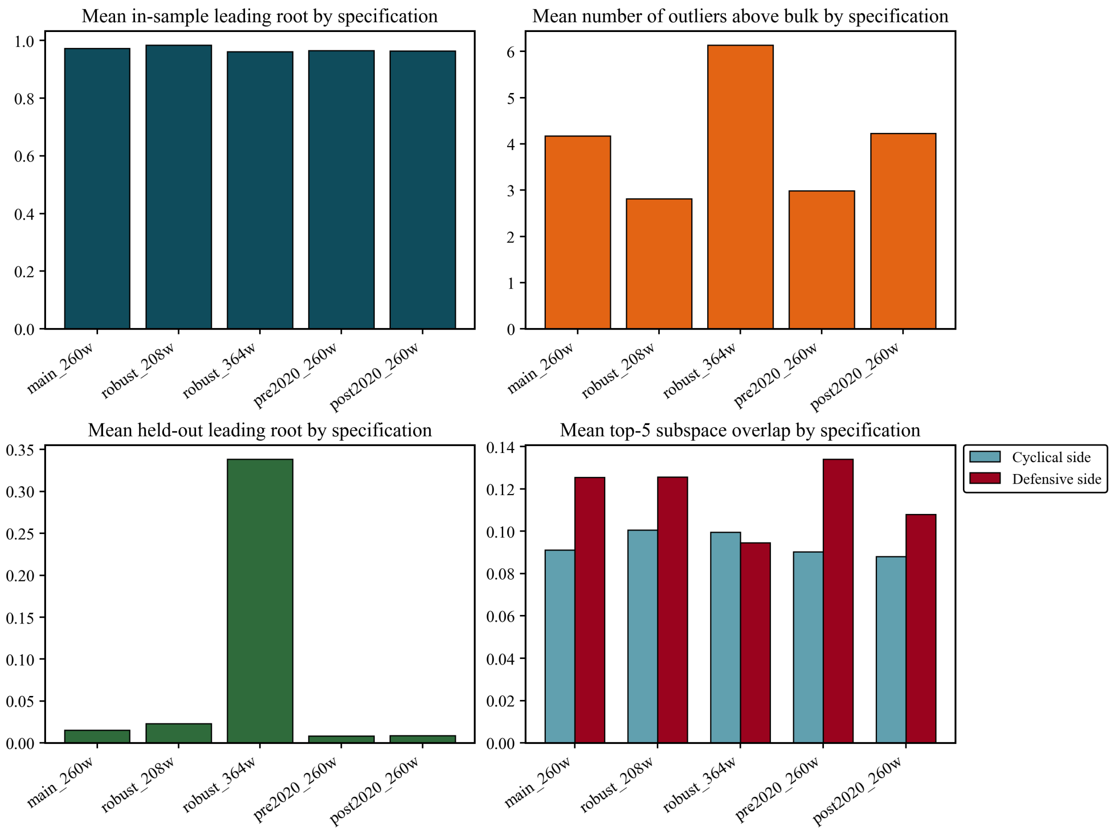


Figure 5.8

Cross-specification robustness comparison for spectral strength, stability, and held-out dependence.



APPENDIX

Appendix A: Monte Carlo Full Results

This appendix reports the full Monte Carlo evidence underlying the main results. The discussion below summarizes the main patterns in the simulations, while the complete tables and figures are collected afterward on a separate page.

Data generation

The Monte Carlo experiments implement a two-panel, low-rank (spiked) covariance model calibrated to the same dimensions later used in the empirical application. The baseline configuration for the data generation process (DGP) is therefore, $p = 80$, $q = 80$, and $n = 520$ observations. Main experiments use 2,000 independent replications whereas the pure-noise and near-threshold designs use 5,000 replications for greater precision in tail and detection-rate estimates. The code implementing these data-generating processes and the associated diagnostics is available in the project repository at https://github.com/albats/hdcca_thesis, with the Monte Carlo notebook located in `Simulations/simulations.ipynb`.

Population cross-covariance is constructed as

$$\Sigma_{xy} = \sqrt{\Sigma_{xx}}\, K\, \sqrt{\Sigma_{yy}}, \qquad K = U_0\, \mathrm{diag}(\rho_1, \ldots, \rho_r)\, V_0^\top,$$

where $\{\rho_j\}_{j=1}^r$ are the population canonical correlations (the spikes) and U_0, V_0 are orthonormal direction matrices. By default $\Sigma_{xx} = I_p$ and $\Sigma_{yy} = I_q$ meaning each panel's variables are standardized and independent in the baseline DGP. Robustness checks then replace these with (i) spiked within-panel covariances to model stronger internal correlation among assets, which is more realistic for financial series; and (ii) heavy-tailed sampling; introducing large outliers and tail risk typical of returns.

For each replication the pipeline is:

1. Construct the joint covariance $\Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{xy}^{\top} & \Sigma_{yy} \end{pmatrix}$;

2. Draw n i.i.d. observations from the chosen law (baseline: multivariate normal; robustness: Student-t with df $= 7$);

3. Compute sample covariance blocks S_{xx}, S_{yy}, S_{xy} from mean-centered data;

4. Whiten the cross-covariance and compute $M = S_{xx}^{-1/2} S_{xy} S_{yy}^{-1/2}$;

5. Obtain the SVD $M = U \operatorname{diag}(\sigma_j) V^{\top}$ and record $\widehat{\lambda}_j = \sigma_j^2$, the estimated canonical weights $\widehat{A} = S_{xx}^{-1/2} U$ and $\widehat{B} = S_{yy}^{-1/2} V$, and all downstream diagnostics.

The recorded metrics per replication include: spectral diagnostics ($\widehat{\lambda}_j$, outlier counts relative to null cutoffs), inflation measures ($\widehat{\lambda}_1 - \rho_1^2$, held-out $\lambda_{1,\mathrm{OOS}}$), geometric measures (principal angles and squared-cosine overlaps between estimated and true canonical directions), and stability measures (split-sample overlaps). These quantities are aggregated across replications to produce the summary tables and figures collected in the Monte Carlo results that follow.

Bykhovskaya-Gorin correction

The BG correction is applied only to sample roots that are separated from the high-dimensional bulk. For each simulated root, the correction inverts the asymptotic mapping between the population canonical root and the observed sample outlier. Roots inside or too close to the bulk are therefore not treated as reliably identifiable population spikes. To account for these, the main tables report the fraction of replications for which the correction produces an admissible separated-root estimate ("Valid BG share").

The corrected value should be interpreted as a de-biased estimate of the population squared canonical correlation, not as an out-of-sample performance measure. The angle predictions provide a separate theoretical benchmark for directional recovery. In the simulations, the correction is most reliable for strong, separated Gaussian spikes and less reliable near the phase-transition boundary or under heavy-tailed sampling.

Experiment 1: Pure-Noise Benchmark

Experiment 1 provides the pure-noise benchmark for the Monte Carlo study. When the two panels are truly independent, the leading sample canonical root ($\hat{\lambda}_1$) is nevertheless large in finite samples, with a mean value of 0.5047, and a standard deviation of 0.0156 (as seen in Table A.1).

This is visually represented in the histogram for the pure noise scenario seen in Figure 3.1 which depicts the leading root centered around a nonzero value despite independence between panels. Such results are reinforced by the scree plot figure Figure A.1 which reflects how, for the pure-noise scenario, there is no dominant root, but rather a smooth bulk-like shape. Experiment 1 makes clear that the empirical leading root must be compared with these thresholds before it can be interpreted as unusually large relative to high-dimensional noise.

The pure-noise design also reveals that the signal captured is unstable as the mean out-of-sample leading root is only 0.0039, while the split-sample squared cosine overlaps are approximately 0.0124 for the X-side vectors and 0.0124 for the Y-side vectors. In simpler terms, the canonical directions extracted under pure noise do not replicate across samples. Taken together, these results imply that a large leading sample root is not, by itself, evidence of meaningful dependence. This experiment therefore provides the finite-sample reference distribution used later to distinguish genuine outliers from noise-generated spikes.

Experiment 2: One-Spike Signal-Strength Sweep

Experiment 2 studies how the leading canonical root reacts when a single true population spike is introduced and its strength is varied. The theoretical high-dimensional threshold is approximately $\rho_c \approx 0.426$, so the key question is whether the sample spectrum begins to separate from the null benchmark before that point. The results, as seen in Figure 3.2, indicate that the leading sample root remains very close to its pure-noise value throughout the weak-signal region. As ρ increases from 0 to 0.345, the mean of $\hat{\lambda}_1$ rises only from 0.5047 to 0.5098, while the second sample root remains essentially unchanged, moving only from 0.4793 to 0.4840. Hence, for subcritical or near-subcritical signal strengths, the sample spectrum is still largely dominated by the high-dimensional noise bulk; see Table A.2

The instability diagnostics reinforce this conclusion. As depicted in Figure A.2, over the same range, the mean out-of-sample leading root stays near zero, fluctuating around 0.004, which is negligible relative to the in-sample value of about 0.50. Likewise, figure A.3 shows that the estimated canonical vectors remain poorly recovered: the average angles are still extremely large, falling only from about 85° to 78°, and the squared cosine overlaps rise only modestly from roughly 0.01 to 0.06. These findings imply that weak population dependence is absorbed into the bulk rather than emerging as a clean, interpretable outlier. Thus, before the signal reaches the threshold region, CCA substantially overstates the strength of dependence in sample while delivering almost no out-of-sample persistence or reliable vector recovery.

Experiment 3: Threshold Zoom

Experiment 3 focuses on the neighborhood of the theoretical high-dimensional threshold, which in this calibration is approximately $\rho_c \approx 0.426$. Here, detection probability measures how often, for a given population canonical correlation ρ, the simulated leading sample root $\widehat{\lambda}_1$ is large enough to exceed the 99% pure-noise cutoff $c_{0.99}$ estimated in Experiment 1. The main result is that detection improves only gradually through this region rather than appearing as a sharp finite-sample jump. When $\rho = 0.306$, the empirical detection probability is only 0.0148; at $\rho = 0.346$ it remains 0.0190; at $\rho = 0.386$ it is still only 0.0326. Even exactly at the theoretical threshold, the detection probability is just 0.0598. Clearer separation only emerges above the threshold where detection rises to 0.1370 at $\rho = 0.466$, to 0.3324 at $\rho = 0.506$, and to 0.6130 at $\rho = 0.546$; see Table A.3. Figure 3.3 depicts the detection probabilities in a visual manner.

Figure A.4 shows that reliable recovery lags behind in-sample spectral separation. The mean leading sample root increases from 0.5087 below the threshold to 0.5531 at the strongest value considered, but the mean out-of-sample root remains extremely small, rising only from 0.0038 to 0.0058 over the same range. At the same time, Figure A.6 depicts how the canonical directions become gradually more stable as the average angle on the X side falls from about 80° at $\rho = 0.306$ to 47° at $\rho = 0.546$, while the squared cosine overlap rises from 0.0439 to 0.4674. Thus, the threshold region should be interpreted as a transition zone rather than a knife-edge boundary. Above ρ_c, outliers become more likely, but vector recovery and out-of-sample persistence still improve only progressively.

Additionally, the BG correction shows that the raw leading roots remain upward biased near the threshold while the BG correction reduces bias once roots separate from the bulk. It should be noted that the fact that below and near the threshold the correction is not as reliable is consistent with the theory, and reinforces the idea that these roots cannot be properly recovered because they are not sufficiently separated.

Experiment 4: Multiple-Spike Low-Rank Design

For experiment 4, three spikes with population canonical correlations $(0.80, 0.55, 0.30)$ are considered. On average, the number of detected outliers is 1.5785, which implies that finite samples typically separate only the strongest part of the low-rank structure. The mean leading three sample roots are 0.7544, 0.5515, and 0.4934, respectively, so the spectrum remains clearly ordered, but only the first two components are meaningfully distinguishable from the noise benchmark; see Table A.4 and Figure A.7.

Recovery reported in Table A.5 confirm this ranking. Recalling that mean angles close to 90° and mean squared cosine overlaps close to zero indicate poor recovery, the results show that the first component is estimated well with a mean angle on the X side of only 20.84° and with mean squared cosine overlap 0.8725. The second component is recovered only moderately well, with mean angle 47.05° and mean squared cosine overlap 0.4692. By contrast, the third component is effectively lost in the bulk as its mean angle is 80.45° and its mean squared cosine overlap is only 0.0407.

Furthermore, the average out-of-sample leading root remains small (at 0.1881), and the mean split-sample squared cosine overlaps are about 0.272 on both sides, again indicating that only part of the latent low-rank structure is stably recovered. Overall, the experiment supports the behavior of HDCCA Bykhovskaya-Gorin formulate, confirming that in a multi-spike design only sufficiently strong spikes emerge as reliable empirical outliers, while weaker ones remain statistically present but practically unrecoverable.

Finally, the component-by-component BG results confirm that correction validity is itself a signal of separation. For the strongest component, with $\rho_1 = 0.80$ and $\rho_1^2 = 0.6400$, the mean raw sample root is 0.7545, while the BG-corrected estimate is 0.6383, with a valid BG share of 1.0000. For the second component, with $\rho_2 = 0.55$ and $\rho_2^2 = 0.3025$, the raw root is 0.5516, but the BG-corrected estimate is 0.2996, again very close to the true value. By contrast, the weakest component, with $\rho_3 = 0.30$ and $\rho_3^2 = 0.0900$, has a raw sample root of 0.4943, but the correction is valid in only 0.0335 of replications and produces an average corrected value of 0.2239. This indicates that the third component is not reliably separated from the bulk, so its apparent sample root should not be interpreted as a well-identified population signal.

Experiment 5: Diffuse Versus Low-Rank Dependence

Experiment 5 focuses on comparing the behavior under a concentrated low-rank dependence structure and a diffuse alternative. That is, a low-rank design where cross-panel dependence is concentrated in a small number of dominant canonical directions. Versus a diffuse design, where the same type of dependence is spread across many weaker directions, making individual components harder to separate from the high-dimensional bulk. Figure A.8 and Figure A.9 compare the spectral shapes produced by each of these designs. The low-rank designs produce a steeper decline at the leading ranks, indicating that part of the dependence is concentrated in a small number of components. By contrast, the diffuse design produces a smoother spectrum with no clear separation point. These results highlight that dependence across several components is not always visible as a few clear outliers. When the signal is spread across many directions, the spectrum changes more gradually and becomes harder to separate from the high-dimensional noise bulk.

Experiment 6: Finance-Style Robustness Checks

Experiment 6 examines whether the main qualitative conclusions survive under more realistic data-generating mechanisms. Three processes are considered while keeping the same one-spike population signal, $\rho = 0.70$, but varying the distributional and covariance assumptions. Specifically, the designs are: (i) the Gaussian identity benchmark, (ii) the Gaussian spiked-covariance design, and (iii) the Student-t identity design. These cases are used as finance-style robustness checks because equity return panels typically exhibit both within-panel dependence, as stocks in the same group tend to co-move, and heavier tails than the Gaussian benchmark, reflecting the presence of large return observations. Holding ρ fixed ensures that differences across the three designs are attributed to covariance structure and tail behavior rather than to changes in the underlying canonical signal.

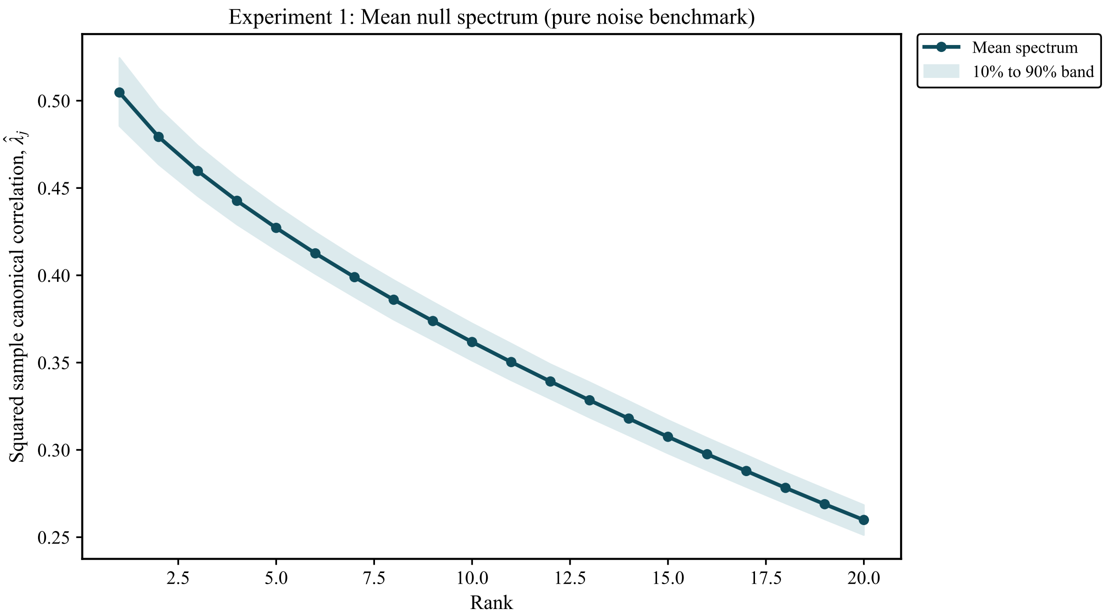
Table A.7 shows that all three finance-style designs produce detection probability equal to one when the population spike is fixed at $\rho = 0.70$. This indicates that the signal remains strong enough to separate from the pure-noise benchmark under each specification. The main differences appear in the stability and recovery measures. The Student-t design has the largest mean in-sample root, but a lower out-of-sample root, lower split-sample stability, larger vector angles, and lower squared cosine recovery than the Gaussian cases. This suggests that heavy-tailed observations can make the in-sample canonical relation appear stronger while making the estimated directions less stable.

Similarly, the BG correction results show that the formulas remain accurate under the clean Gaussian designs but become less reliable under heavy-tailed sampling. In the Gaussian identity benchmark, where the true value is $\rho^2 = 0.4900$, the mean raw leading root is 0.6582, while the BG-corrected estimate is 0.4873, giving a negligible bias of -0.0027. The Gaussian spiked-covariance design gives almost identical results; the raw root is 0.6580, the corrected estimate is 0.4871, and the correction bias is -0.0029. In both cases, the valid BG share is 1.0000. Under Student-t sampling, however, the raw root rises to 0.7138, and the BG-corrected estimate remains 0.5755, implying an upward bias of 0.0855. Given that the instability diagnostics move in the same direction, this suggests that the heavy tails do not prevent detection in this calibration, but they make both the raw root and the BG-corrected root look stronger than the recoverable canonical direction actually is.

Metric	Value
Mean $\hat{\lambda}_1$	0.5047
Standard deviation of $\hat{\lambda}_1$	0.0156
95% null cutoff for $\hat{\lambda}_1$	0.5312
99% null cutoff for $\hat{\lambda}_1$	0.5454
Mean out-of-sample $\hat{\lambda}_1$	0.0039
Mean split stability, X	0.0124
Mean split stability, Y	0.0124

Table A.1
Experiment 1 summary statistics under the pure-noise benchmark.

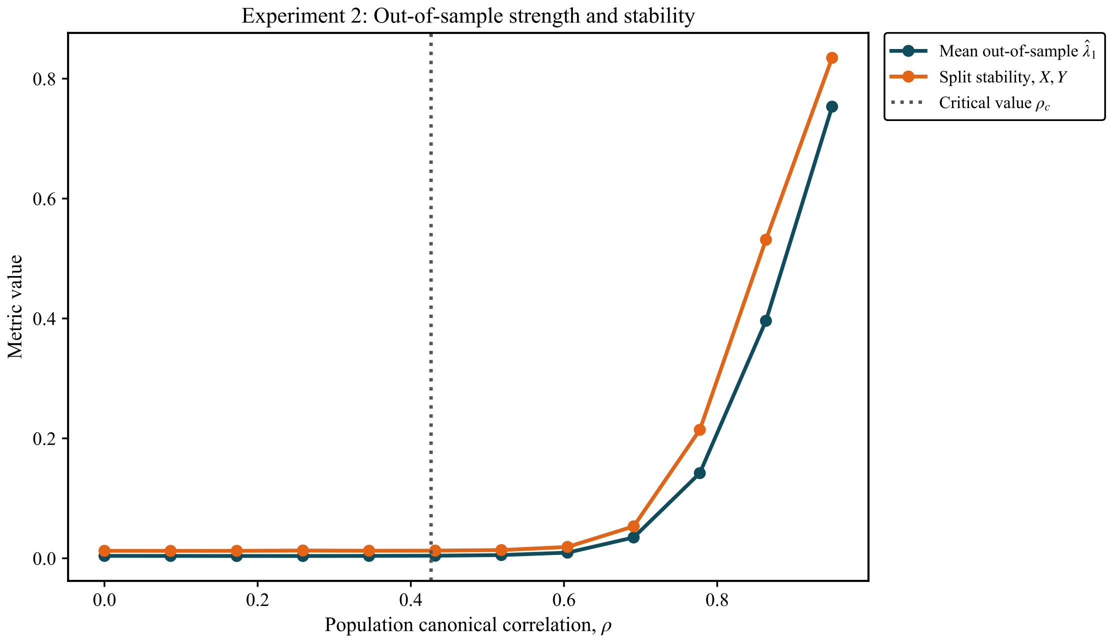
Figure A.1
Scree plot for the mean canonical roots for the pure-noise scenario (Experiment 1). The scree plot figure displays the ordered canonical roots, $\widehat{\lambda}_1 \geq \widehat{\lambda}_2 \geq \cdots$, where $\widehat{\lambda}_k = \widehat{\rho}_k^2$ (the rank *in the x-axis refers to the position of each canonical root after sorting them from largest to smallest) and it can be used to assess whether the leading canonical roots separate from the rest of the spectrum. In the pure-noise scenario it displays a smooth, bulk-like structure.*



ρ	ρ^2	Mean $\hat{\lambda}_1$	Mean $\hat{\lambda}_2$	Mean out-of-sample $\hat{\lambda}_1$	Mean split stability, X	Mean split stability, Y	Mean angle, X (deg)	Mean angle, Y (deg)
0.0000	0.0000	0.5047	0.4793	0.0040	0.0125	0.0127	84.9259	85.0234
0.0864	0.0075	0.5051	0.4790	0.0039	0.0123	0.0126	84.6441	84.5019
0.1727	0.0298	0.5055	0.4801	0.0038	0.0124	0.0125	83.6085	83.8085
0.2591	0.0671	0.5073	0.4817	0.0038	0.0130	0.0125	81.7284	81.8881
0.3455	0.1193	0.5096	0.4834	0.0040	0.0126	0.0124	77.9253	77.8977
0.4318	0.1865	0.5187	0.4882	0.0043	0.0128	0.0118	68.4277	68.4388
0.5182	0.2685	0.5405	0.4947	0.0054	0.0137	0.0136	52.4511	52.5062
0.6045	0.3655	0.5859	0.4976	0.0094	0.0189	0.0192	38.4334	38.5030
0.6909	0.4774	0.6510	0.4992	0.0347	0.0533	0.0540	29.3181	29.3359
0.7773	0.6042	0.7311	0.4995	0.1420	0.2143	0.2130	22.1970	22.2506
0.8636	0.7459	0.8256	0.4993	0.3961	0.5314	0.5298	15.7990	15.8161
0.9500	0.9025	0.9325	0.5004	0.7533	0.8348	0.8345	8.8547	8.8567

Table A.2
Experiment 2 summary table for the one-spike signal-strength sweep.

Figure A.2
Experiment 2 out-of-sample strength and split-sample stability. The figure reports the held-out leading canonical root and the split-sample directional overlaps. Both diagnostics remain close to zero for weak signals and become substantial only for sufficiently strong population canonical correlations, showing that in-sample separation alone does not guarantee stable or generalizable canonical structure.



ρ	Detection probability	Mean $\hat{\lambda}_1$	Mean out-of-sample $\hat{\lambda}_1$	Mean split stability, X	Mean split stability, Y	Mean angle, X (deg)	Mean angle, Y (deg)
0.3064	0.0134	0.5081	0.0039	0.0120	0.0128	80.0161	80.0453
0.3464	0.0196	0.5100	0.0040	0.0125	0.0126	77.4099	77.3580
0.3864	0.0282	0.5128	0.0040	0.0131	0.0129	73.7812	73.8870
0.4264	0.0606	0.5172	0.0042	0.0131	0.0123	68.9269	68.9236
0.4664	0.1380	0.5245	0.0045	0.0129	0.0129	62.4521	62.5919
0.5064	0.3264	0.5368	0.0048	0.0132	0.0134	54.6496	54.6917
0.5464	0.5982	0.5525	0.0058	0.0148	0.0149	47.1674	47.1475

Table A.3
Experiment 3 summary table around the near-critical region.

Figure A.3

Experiment 2 geometric recovery of the leading canonical vectors. *The figure reports the angle between the estimated and the true leading canonical directions for the X- and Y-side panels. Angles close to* 90° *indicate little directional recovery. Recovery improves only once the population signal moves above the critical region (represented by the vertical line).*

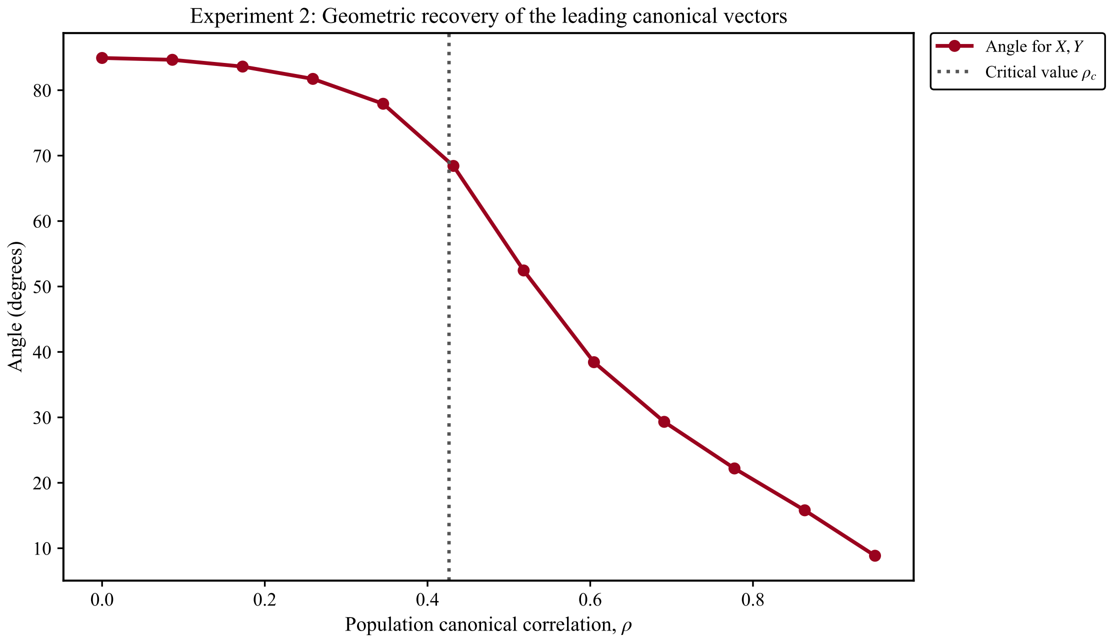


Figure A.4

Experiment 3 in-sample and out-of-sample strength near the threshold. *The figure compares the mean in-sample leading root with the mean out-of-sample leading root. The in-sample root remains close to the high-dimensional pure-noise cutoff, while the out-of-sample root stays close to zero, showing that near-threshold signals do not necessarily produce stable held-out canonical correlation.*

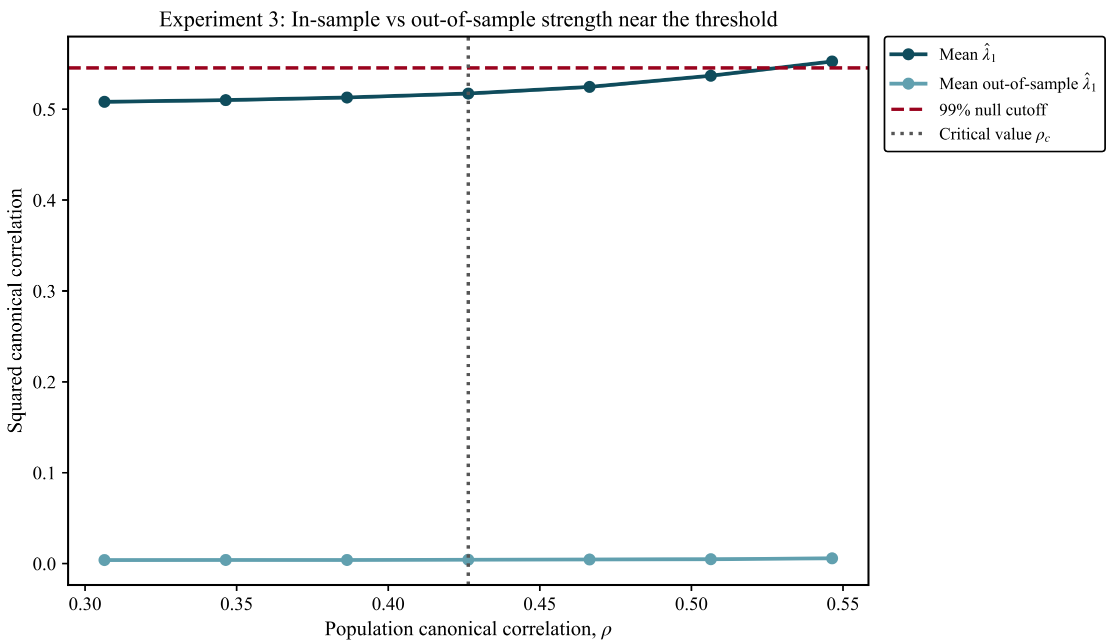


Figure A.5

Experiment 3: BG-predicted outlier map around the threshold. The figure reports the predicted leading root from the Bykhovskaya-Gorin formulas.

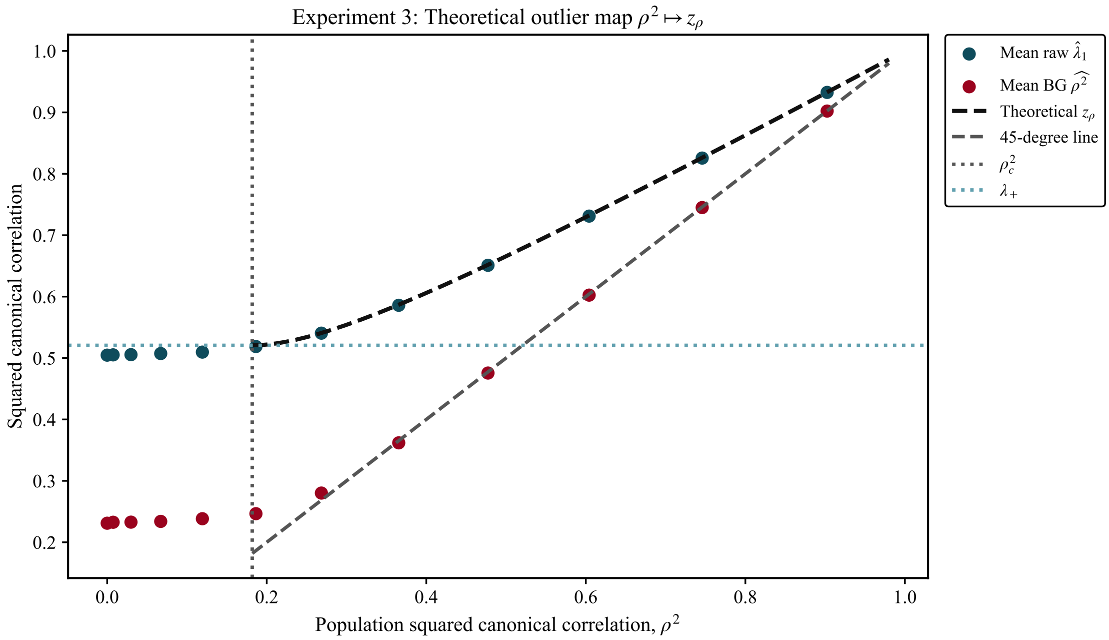
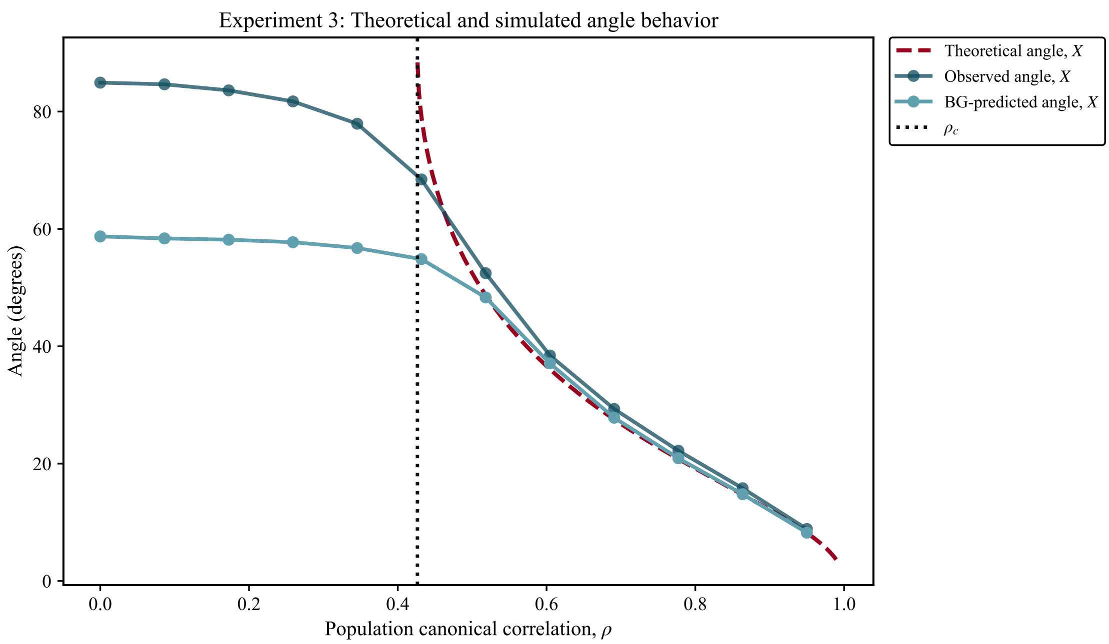


Figure A.6

Experiment 3: Theoretical and simulated angle behavior around the threshold. The figure reports the observed mean angle between the observed and BG-predicted angles. In both cases angles decline as the population canonical correlation ρ increases, but remain large near the critical value ρ_c, indicating weak directional recovery in the near-threshold region. Additionally, the BG-predicted angles are close to the theoretical predictions.



Metric	Value
Mean number of outliers	1.5885
Mean out-of-sample $\hat{\lambda}_1$	0.1894
Mean split stability, X	0.2756
Mean split stability, Y	0.2754
Mean $\hat{\lambda}_1$	0.7545
Mean $\hat{\lambda}_2$	0.5516
Mean $\hat{\lambda}_3$	0.4943
Mean angle, X, comp. 1 (deg)	20.8568
Mean angle, X, comp. 2 (deg)	47.2661
Mean angle, X, comp. 3 (deg)	80.4800
Valid BG share, comp. 1	1.0000
Valid BG share, comp. 2	0.9020
Valid BG share, comp. 3	0.0335
Mean BG $\widehat{\rho_1^2}$	0.6383
Mean BG $\widehat{\rho_2^2}$	0.2996
Mean BG $\widehat{\rho_3^2}$	0.2239

Table A.4
Experiment 4 summary statistics for the three-spike low-rank design.

Component	ρ_j	ρ_j^2	Mean $\hat{\lambda}_j$	Mean BG $\widehat{\rho_j^2}$	Mean θ_X	Mean BG θ_X	Mean squared cosine, X
1	0.80	0.64	0.75	0.64	20.86	19.27	0.87
2	0.55	0.30	0.55	0.30	47.26	45.11	0.47
3	0.30	0.09	0.49	0.22	80.48	60.59	0.04

Table A.5
Experiment 4 component-by-component recovery measures.

Design	Signal rank	Nº outliers	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\lambda}_3$	OOS $\hat{\lambda}_1$	Split stability, X	Valid BG share, comp.1	BG $\widehat{\rho_1^2}$	BG θ_X, comp. 1	θ_X, comp.1
Diffuse	10	0.1300	0.5274	0.5003	0.4801	0.0048	0.0127	0.6530	0.2521	53.4246	78.7302
Low-rank	3	2.1085	0.7349	0.5944	0.5217	0.1397	0.1901	1.0000	0.6083	20.6392	23.0050

Table A.6
Experiment 5 comparison of the low-rank and diffuse designs. Note that the table reports mean values across simulations.

Figure A.7

Experiment 4 mean scree plot under the three-spike low-rank design. The figure reports the mean ordered sample canonical roots. The spectrum displays a steep decline at the leading ranks followed by a smoother decay across the remaining roots. This pattern is consistent with dependence concentrated in a small number of canonical directions, although the lower-ranked components merge progressively into the high-dimensional bulk.

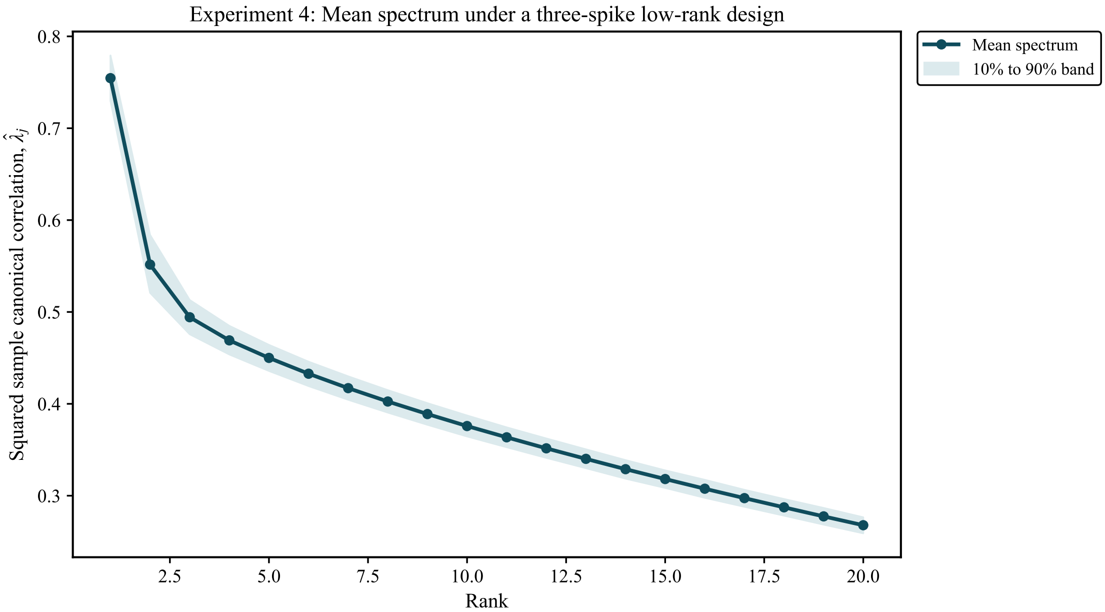


Figure A.8

Experiment 5A mean scree plot under the low-rank design. The figure reports the mean ordered sample canonical roots where the leading roots are visibly larger than the rest of the spectrum, after which the curve declines smoothly. This indicates that the simulated dependence is concentrated in a limited number of directions.

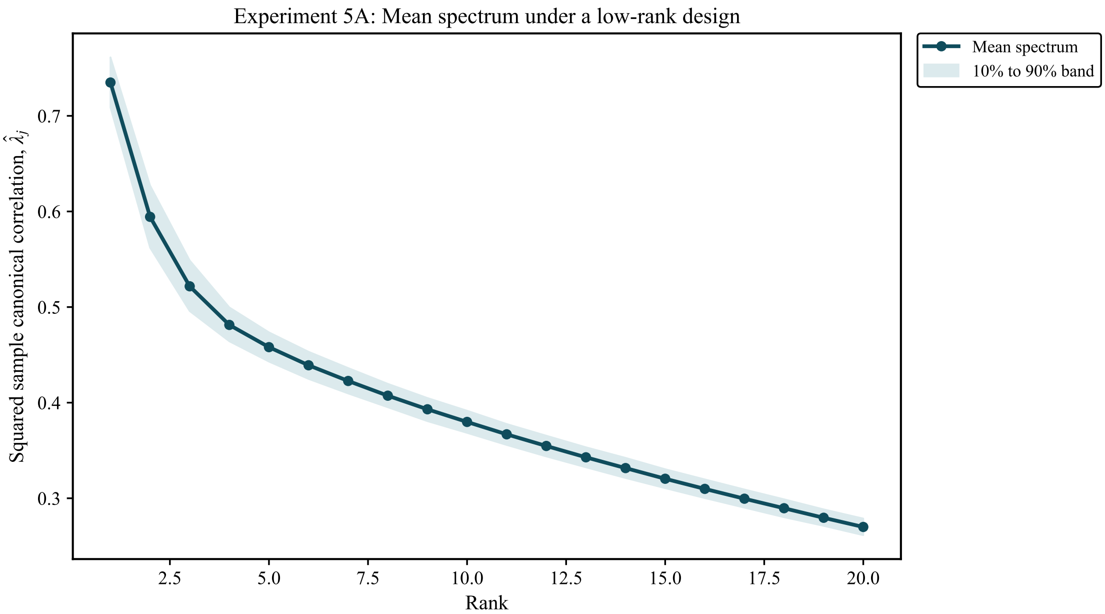
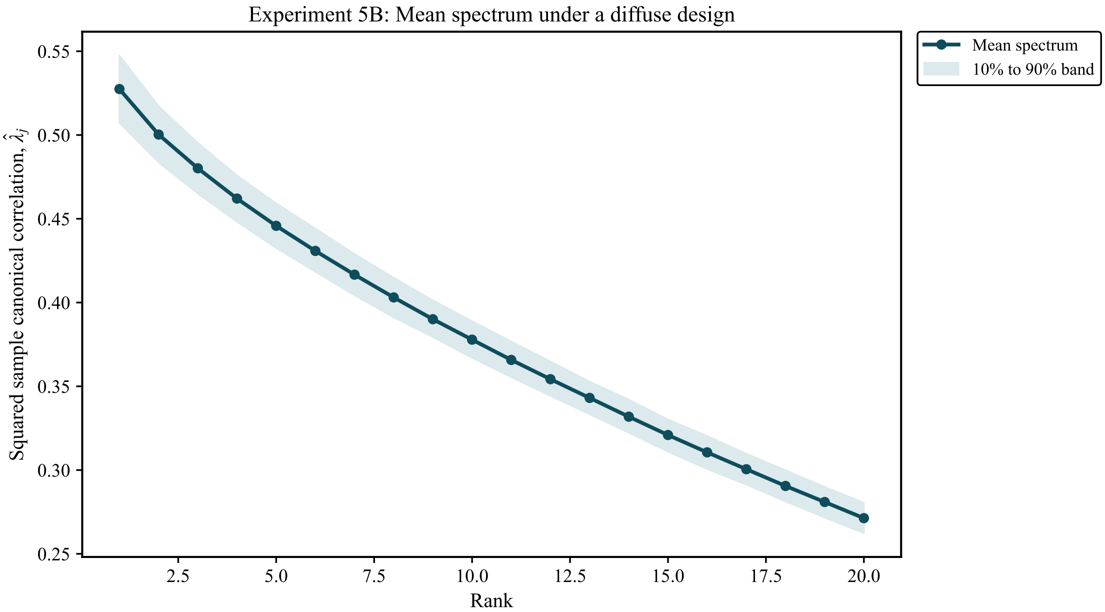


Figure A.9
Experiment 5B mean scree plot under the diffuse design. The figure reports the mean ordered sample canonical roots. Unlike the low-rank case, the spectrum declines gradually without a clear break between leading and non-leading roots. This shows that diffuse dependence is harder to identify through isolated sample outliers, since signal strength is spread across several directions rather than concentrated in a small number of dominant roots.



Setting	$\hat{\lambda}_1$	OOS $\hat{\lambda}_1$	BG $\widehat{\rho^2}$	BG θ_X (deg)	Split stability	θ_X (deg)	Squared cosine
Gaussian identity	0.66	0.04	0.49	27.08	0.06	28.59	0.77
Gaussian spiked covariance	0.66	0.04	0.49	27.09	0.06	28.61	0.77
Student-t identity	0.7138	0.03	0.58	22.26	0.04	35.34	0.67

Table A.7
Experiment 6 summary statistics across the robustness designs. The table reports the main Monte Carlo diagnostics for three data-generating processes that keep the same one-spike population signal, $\rho = 0.70$, but vary the distributional and covariance assumptions. Note that the table reports mean values across simulations. Additionally we only report angle X as p=q=80 makes the angles on both sides very similar. Specifically, the designs include (i) the Gaussian identity benchmark, (ii) the Gaussian spiked-covariance design, (iii) and the Student-t identity design. Detection probability is equal to one in all three settings, showing that the leading root remains clearly distinguishable from the pure-noise benchmark. However, the recovery diagnostics differ across designs. The Student-t identity case produces the largest in-sample root, but also the weakest out-of-sample root, the lowest split-sample stability, and the largest average angles between estimated and true canonical directions. Thus, heavy tails do not prevent detection in this calibration, but they reduce the reliability of the estimated canonical vectors.

Appendix B: Empirical Application Supplementary Results

Item	Value
Sample span	2010-01-08 to 2025-12-12
Total weekly observations	832
Cyclical assets (p)	80
Defensive assets (q)	80
Rolling window length	260 weeks
Step size	4 weeks
Number of windows	144
Window bulk edge λ_+	0.8521
Critical population threshold ρ_c^2	0.4444
Coverage threshold	0.99

Table B.1
Baseline CRSP sample design used in the empirical rolling-window application.

Rank	Squared root	Above bulk edge
1	0.9389	Yes
2	0.7588	Yes
3	0.6457	Yes
4	0.5956	Yes
5	0.4834	Yes
6	0.4583	Yes
7	0.4191	Yes
8	0.3920	Yes
9	0.3754	Yes
10	0.3606	Yes

Table B.2
Top ten squared canonical roots from the full-sample static spectrum, with the HDCCA bulk-edge classification.

Figure B.1

Full-sample static canonical spectrum for the CRSP cyclical and defensive panels.

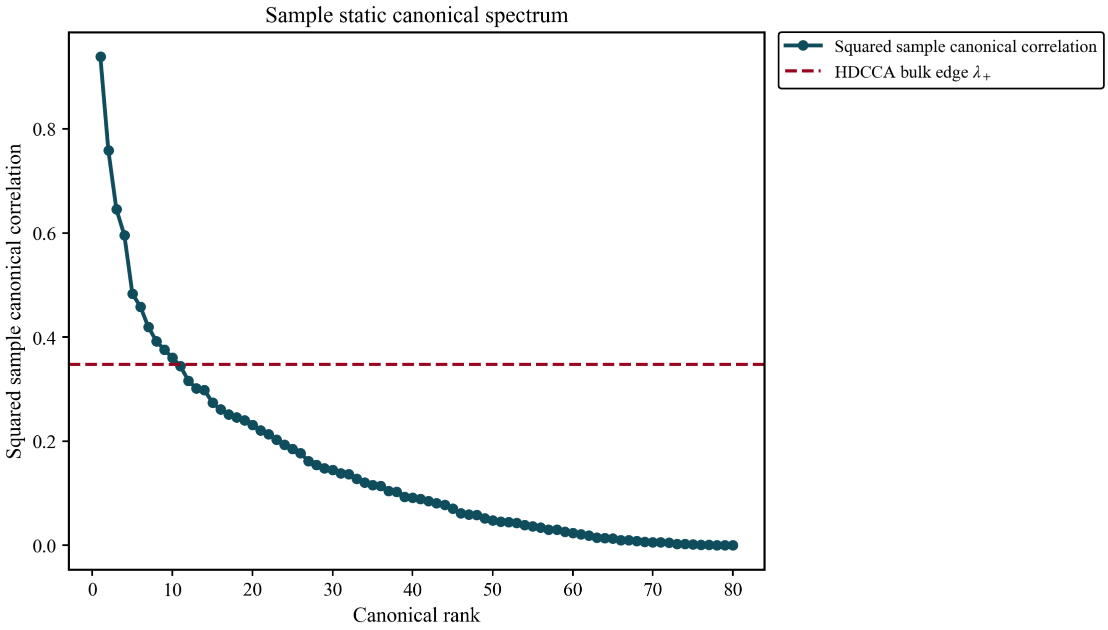


Figure B.2

Full-sample canonical score scatter plots for the first three outlying canonical signals. The first pair of canonical scores is tightly concentrated around the diagonal, indicating a very strong leading common component between the cyclical and defensive panels. The second and third score pairs remain positively associated but are visibly more diffuse, suggesting weaker additional canonical signals. This pattern is consistent with the Bykhovskaya-Gorin interpretation of a strong leading factor accompanied by weaker, less sharply separated additional factors.